

Global Fisheries: Quantifying the Externalities from Open Access

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Abstract

This paper uses a novel geospatial dataset of global fishery catch and develops a quantitative dynamic spatial model to quantify the externalities from open access in global fishing. I first show that (i) the average global fishery stock decreased by 35% between 1980 and 2018, (ii) assigning property rights lead to larger fishery stock, and (iii) fuel subsidies to vessels are positively correlated with high sea fishing. Then, I build a dynamic spatial model of global fisheries and compare two polar cases of open access: the decentralized equilibrium, where atomistic firms have open access to the fishery, and the socially optimal allocation, where the social planner has exclusive property rights. By taking the model to the data in 2018, I find that in the socially optimal allocation, the average global fishery stock at the steady state increases by 88%, and the net present value of global welfare increases by 0.11%, compared to the decentralized equilibrium. The counterfactual analysis shows that fuel subsidies are globally welfare-reducing and decrease the average global fishery stock at the steady state by 3.2%.

Keywords: International trade, natural resources, global fishery, tragedy of commons

JEL Codes: F13, F18, H23, Q22

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1 Introduction

The tragedy of the commons, a market failure to internalize the social cost under open access, has long been studied in economics (Hardin, 1968; Ostrom, 1990), but quantifying its economic impact remains challenging. Global fisheries present a prominent example of the tragedy of commons. Since 1980, the proportion of over-fished species has risen from 20% to 35%, underscoring the increasing pressure on marine resources (FAO, 2022). Fish are not only essential for marine biodiversity but also play a critical role in human diets, accounting for 20% of total animal protein intake globally (FAO, 2024b). For some developing countries, the share of fish in total animal protein intake is much higher, for example 52% in Indonesia, emphasizing its significance in food security.¹ Recognizing the importance of fisheries, multilateral global initiatives have sought to preserve fishery stocks by establishing the Exclusive Economic Zones (EEZs) or designating protected areas (UN, 1982, 2023).² Despite these efforts, individual countries unilaterally continue to subsidize fishing activities (Sala et al., 2018).

This paper quantifies the externalities in global fisheries arising from open access and investigates how global fishery policies impact the spatial distribution of fishery activities and global welfare. I compile a novel geospatial dataset on fishery catch, fishery stock, and international trade. Guided by the stylized facts about the patterns of global fisheries with open access and subsidies, I develop a dynamic spatial model of global fisheries featuring multiple fishing locations, exporters, and importers, along with the evolution of fish stocks. To quantify the externalities from open access, I compare the outcomes from two polar cases: a decentralized equilibrium, where atomistic firms have open access to the fishery, and the optimal allocation, where a social planner has exclusive property rights. I analytically characterize the difference in outcomes between the two, where the social planner fully internalizes the externalities of fishery production while the atomistic firms do not. The quantitative results show that in the social planner allocation, the average global fishery stock at the steady state increases by 88%, and the net present value of global welfare increases by 0.11%

¹The reliance on fish as a protein source varies substantially across countries, ranging from Indonesia’s 52% to Japan’s 34%, China’s 25%, and the USA’s 7% (FAO, 2024b).

²A landmark initiative was the declaration of Exclusive Economic Zones (EEZs) in 1994 under the United Nations Convention on the Law of the Sea. By providing coastal nations with property rights over marine resources within 200 nautical miles of their shores, the EEZ framework aimed to foster responsible management of marine resources (UN, 1982). More recently, attention has shifted to preserving fish stocks beyond national jurisdiction. The 2023 High Seas Treaty aims to protect 30% of the global ocean through the designation of marine protected areas (UN, 2023).

compared to the decentralized equilibrium. To understand policy impacts, I conduct a counterfactual analysis of permanently eliminating fuel subsidies and find that fuel subsidies are globally welfare-reducing and decrease the average global fishery stock at the steady state by 3.2%.

I begin by compiling a novel comprehensive geospatial dataset, consisting of global fishery catch, global fishery stocks, international trade, and other complementary statistics. First, the geospatial dataset on global fishery catch from Sea Around Us provides detailed information on catch weights by countries and species at the grid-level ($0.5^\circ \times 0.5^\circ \approx 60km \times 60km$). Second, the fishery stocks are estimated at the grid level using the estimation package from natural science literature, taking the fishery catch data as one of the inputs. Third, I construct a fishery trade dataset with domestic consumption by harmonizing bilateral fishery trade flows with FAO food balance sheets. Fourth, I collect other complementary statistics on fishery such as fishery habitat suitability, ex-vessel prices, employment, and subsidies, which I use for model calibration and estimation. By combining the geospatial catch dataset (“production flows”) with the international trade dataset (“trade flows”), I track the flow of fisheries from fishing locations to final destinations, offering a valuable resource for quantitative analysis.

Using this dataset, I present three empirical findings that motivate my quantitative model of global fisheries. First, in the past 40 years, global fishery stocks have decreased by 35% on average, and the number of overfished species has more than doubled. Second, the assignment of property rights is associated with larger fishery stock. Taking the declaration of EEZ in 1994 as a quasi-experiment, I show that the fishery stock increased in the EEZ relative to the high sea after the declaration. Third, government subsidies to the fishing industry, such as fuel subsidies toward vessels, are correlated with high seas fishing. This finding suggests that understanding the spatial dimension of the global fishery is crucial to examining the impact of fishery policies. In addition, this finding is informative about the key elasticity of substitution across fishing locations at the calibration.

I then develop a dynamic spatial model of global fisheries that incorporates the evolution of fishery stocks under a global trade system. The quantitative model accounts for multiple exporters and importers, multiple fishing locations, and two industries with outside good and fishery sectors. The fishery output is consumed as nested constant elasticity of substitution (CES), where fish are differentiated by the order of species, exporters, and fishing locations at each tier of the nests.³ The fishery stocks evolve according to a law of motion governed by ecological growth function and anthropogenic activity, closely following the literature

³An example of differentiation across fishing locations at the bottom tier is the Tuna from the Pacific Ocean caught by China versus the Tuna from the Atlantic Ocean caught by the USA.

from natural science on population dynamics. Firms in the fishery sector are subject to two frictions: iceberg commuting costs to reach fishing locations from home countries and iceberg trade costs to ship fish from home countries to final destinations.

The quantitative model features two key characteristics. First, fishing productivity is an increasing function of fishery stocks. Fishery production today results in the fishery sector becoming less productive in the future, implying an additional cost of fishing. I refer the loss of future productivity from fishing as the dynamic social cost. Second, the only dynamic force in the decentralized equilibrium is the evolution of fishery stock. The atomistic firms ignore the impact of harvesting on future stock and solve a static optimization problem, not accounting for the dynamic social cost. As a result, the decentralized equilibrium allocation equalizes the marginal utility of fishery consumption with the static marginal cost of fishing, which is the input cost per unit of output.

To quantify the externalities, I compare the decentralized equilibrium with the optimal allocation by the social planner. I begin with characterizing the planner's problem that directly chooses the allocation of labor and consumption. In contrast to atomistic firms in the decentralized equilibrium, the planner fully internalizes the dynamic social cost of fishery production. From the planner's first-order conditions, I show that the planner allocation equalizes the marginal utility of fishery with the total cost of fishing, which includes both the static marginal cost and the dynamic social cost. The dynamic social cost of fishing that the decentralized equilibrium fails to internalize becomes the source of the externality. Since the dynamic social cost is positive and the marginal utility of fishery consumption diminishes, the planner allocates fewer inputs to the fishery and keeps a larger fishery stock than in the decentralized equilibrium.

Then, I take the quantitative model to data. I disaggregate the world into 30 countries, comprising 25 individual countries and 5 continental groups, chosen based on the size of fishery industries and overall economies. Additionally, I divide the high seas into 16 regions as defined by FAO fishing regions, resulting in a total of 46 fishing locations. I group fish species into 10 broader species following the ISSCAAP classification. By calibrating the model to data in 2018, I estimate the elasticities of substitution from IV regression that uses the exogenous variations in the geographical proximity to fishery habitat as a supply-side shock. Finally, I recover the fundamentals from model inversion.

I use the calibrated model to quantitatively address the two questions of this paper. First, I compare the baseline decentralized equilibrium with the optimal allocation by the social planner. This comparison allows me to measure the inefficiencies associated with the externalities in global fisheries. Second, I examine a counterfactual scenario where I permanently eliminate all fuel subsidies. This exercise allows me to quantify the effect of

fishery policy on the spatial distribution of fishing activities and global welfare.

In the baseline decentralized equilibrium, I find that the average global fishery stock decreases by 32% at the steady state compared to the initial year of 2018. With the fishery stock becoming scarce, the fishery sector becomes less productive and the fish becomes more expensive, resulting in lower fishery consumption at the steady state. In contrast, in the socially optimal allocation, the average global fishery stock increases by 27% at the steady state compared to the initial year of 2018. Comparing steady states, the average global fishery stock in the socially optimal allocation is 88% larger than the average global fishery stock in the decentralized equilibrium. In the socially optimal allocation, the net present value of global welfare increases by 0.11% compared to the decentralized equilibrium. The reason is that the planner internalizes the effect that fishing today has on the productivity of fishing tomorrow, and consequently allocates fewer workers toward fishery than in the decentralized equilibrium. While less fish is consumed in the earlier periods, fishery consumption exceeds the level of the decentralized equilibrium over time since the accumulated stock increases fishery productivity and allows for more fishery consumption even with less labor. Moreover, since more labor is allocated to the outside good sector, the consumption of outside good is larger than in the decentralized equilibrium.

Lastly, I turn to examine the impact of fuel subsidies. In a scenario where all such subsidies are permanently eliminated, I find the average global fishery stock at the steady state increases by 3.2% compared to the steady state of the baseline. The net present value of global welfare increases by 0.004%. While the magnitude is small, the result implies that fuel subsidies are welfare-reducing from a global perspective. By eliminating fuel subsidies, as in the case of the planner, fishery consumption decreases in the earlier periods but increases as the stock accumulates.

Related Literature. This paper contributes to three major strands of literature. First, this study contributes to the growing body of work at the intersection of trade, spatial economics, and the environment. [Copeland and Taylor \(2004\)](#) and [Copeland et al. \(2022\)](#) provide comprehensive overviews. Recent advances, as noted by [Desmet and Rossi-Hansberg \(2024\)](#), have incorporated spatial dimensions into quantitative models. Several recent studies have focused on climate change and air pollution ([Desmet and Rossi-Hansberg, 2015](#); [Costinot et al., 2016](#); [Shapiro, 2016](#); [Gouel and Laborde, 2021](#); [Conte et al., 2021](#)). Recent work has also started studying the quantitative relationship between trade and natural resources. These papers include [Farrokhi \(2020\)](#) on global oil markets, [Dominguez-Iino \(2023\)](#) on environmental policies in South American supply chains, [Farrokhi et al. \(2024\)](#) on global deforestation, [Hsiao \(2024\)](#) on international cooperation in the palm oil market and [Carleton et al. \(2024\)](#) on the agricultural trade and the spatial allocation of global water use. My

paper is unique in characterizing the optimal allocation by the planner and quantifying the externalities from open access to natural resources in the context of global fisheries. To my knowledge, this is the first paper to apply a dynamic spatial quantitative model to global fisheries.

Second, I build upon the literature examining optimal environmental policies in open economies. Early theoretical work includes [Markusen \(1975\)](#), [Copeland \(1996\)](#), and [Hoel \(1996\)](#). Recent studies by [Kortum and Weisbach \(2021\)](#) and [Weisbach et al. \(2023\)](#) analyze unilaterally-optimal carbon tax policies in two-country models. [Elliott et al. \(2010\)](#) and [Shapiro \(2021\)](#) develop quantitative analyses of environmental policies. More recently, [Farrokhi and Lashkaripour \(2024\)](#) characterize optimal climate and trade policies in a multi-country general equilibrium model and quantitatively assess the performance of these policies. The contribution of my paper lies in the quantitative analysis of optimal allocation in dynamic settings.

Lastly, I contribute to the extensive literature on the extraction of natural resources and fisheries. Building on the seminal work by [Hotelling \(1931\)](#), natural resource extraction has been widely studied ([Chichilnisky, 1994](#); [Brander and Taylor, 1997, 1998](#); [Copeland and Taylor, 2009](#); [Anderson et al., 2018](#); [Arkolakis and Walsh, 2023](#); [Noack and Costello, 2024](#)). While early works on fisheries were primarily theoretical ([Gordon, 1954](#); [Smith, 1969](#)), recent empirical studies have focused on specific regions or species ([Costello and Polasky, 2008](#); [Huang and Smith, 2014](#); [Kroodsma et al., 2018](#); [Fenichel et al., 2020](#)). I complement these studies by linking the global fishery catch data with international trade and offering a global general equilibrium quantification of externalities from open access to fishery.

The rest of the paper proceeds as follows. Section 2 describes the data sources used in the paper. Section 3 establishes empirical facts to motivate the main framework in the paper. Section 4 builds the dynamic spatial model of global fishery. Section 4.3 describes the planner’s problem and discusses the comparison with the decentralized equilibrium. Section 5 calibrates the model by taking the model to data. Section 6 shows the quantitative results from the decentralized equilibrium, the social planner, and the counterfactual scenario. Section 7 concludes.

2 Data

This section describes a novel dataset that combines geospatial datasets of global fishery catch and stock with international trade flows and other fishery datasets.

Fishery Catch. The fishery catch dataset is sourced from Sea Around Us.⁴ The dataset spans annually since 1980 for approximately 200 countries, covering 1,100 species worldwide. This dataset provides the catch quantity in tonnes for each species by individual country at the grid level of fishing location ($0.5^\circ \times 0.5^\circ \approx 60km \times 60km$). The grid level estimation process integrates information from several sources, including grid-level natural habitat conditions, national statistics by individual countries, and the Food and Agriculture Organization fishery database (Zeller and Pauly, 2015).⁵ To give a graphical illustration of the catch dataset, Figure 1 panel (a) displays a map of the catch of Big Eye Tuna in 2018.

Fishery Stock. I employ the CMSY++ package provided by the Sea Around Us to estimate fishery stock levels. The package takes historical catch data and biological information of species as inputs and estimates the time series of fishery stock.⁶ The estimation also allows me to infer the carrying capacity, which is the maximum population of species that can be attained in the given geography. The carrying capacity suggests the level of fishery stock that would be attained without any anthropogenic activity. Taking the inputs as given, the model infers the stock and carrying capacity that best rationalizes the time series of observed catch.⁷ The machine learning algorithm, trained on direct observation data, enhances the estimation accuracy.⁸ Inputting the fishery catch dataset to the CMSY++ package, I obtain the time series of fishery stock since 1980 and the carrying capacity for 1,100 species at the grid-level. To give a graphical illustration of the stock dataset, Figure 1 panel (b) displays a map of the stock of Big Eye Tuna in 2018.

Fishery Trade. I construct a fishery trade dataset by combining bilateral trade flows with domestic consumption, both sourced from FAO. The FAO dataset on fishery trade provides the bilateral trade flows, both in quantity and value, between 240 countries for over 1,000

⁴Sea Around Us is the research initiative at the University of British Columbia, specializing in quantifying the impact of fisheries on marine environments.

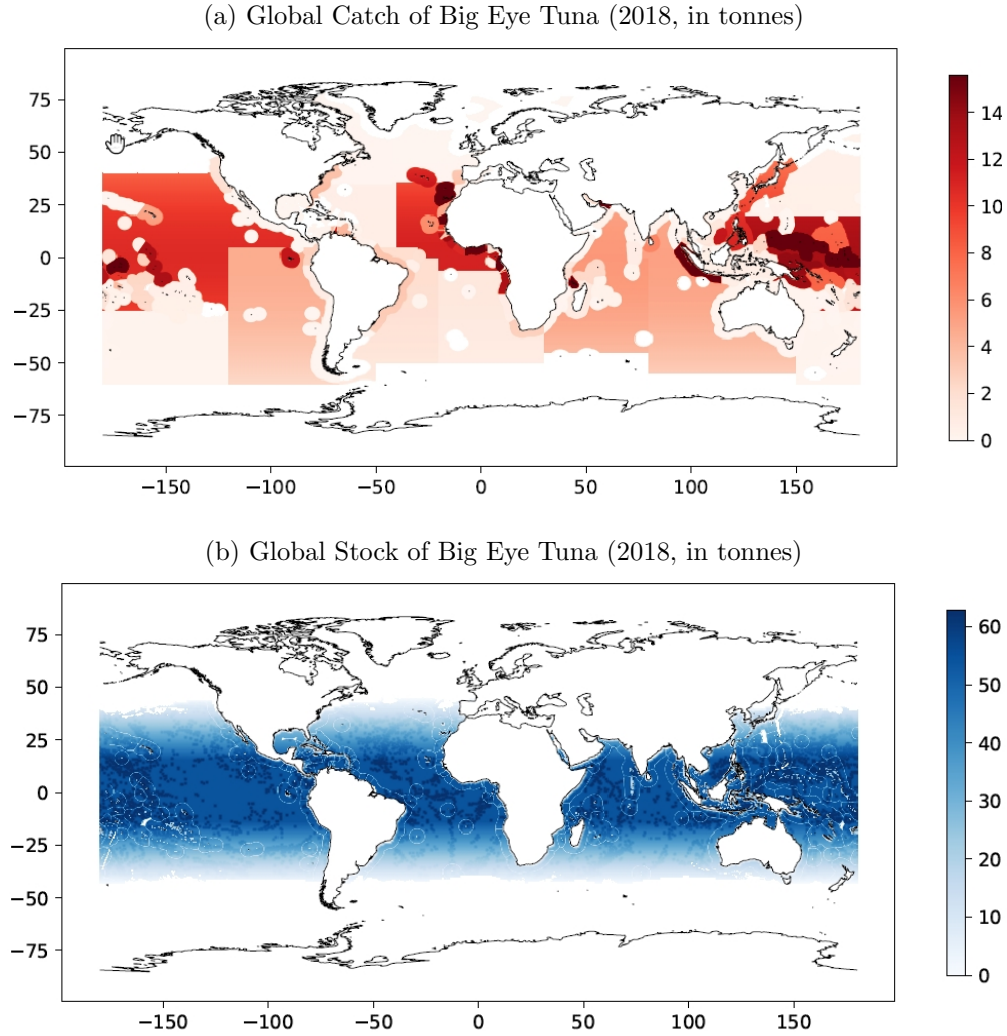
⁵The FAO publishes the annual global fisheries statistics with fishing location information based on individual country surveys (FAO, 2024a). However, the FAO dataset’s geographic units are much broader than the Sea Around Us dataset, classifying the global ocean into 19 FAO major fishing areas. The primary benefit of the granular dataset from Sea Around Us is that it allows me to distinguish between EEZ and high sea.

⁶Estimating fishery stock is inherently more complicated than estimating fishery catch. Broadly, there are two methods of measuring fishery stock in the literature: direct estimation via research vessels, which is limited in scope and commonly conducted by developed countries, and indirect estimation from historical catch and biological data. The indirect method, while broader and longitudinal, relies on assumptions about the model of stock growth. CMSY++ package combines the indirect method with a machine learning algorithm.

⁷The Schaefer growth model from CMSY++ is consistent with the growth model used in this paper.

⁸Froese et al. (2023) validate this methodology, demonstrating that stock levels for 91% of species in the training set fall within the model’s 95% confidence interval in cross-validation.

Figure 1: Global Catch and Stock of Big Eye Tuna (2018, in tonnes)



Notes: Panel (a) shows the spatial distribution for the global catch of Big Eye Tuna in 2018. Panel (b) shows the spatial distribution for the global stock of Big Eye Tuna in 2018.

products since 2019.⁹ Dealing with domestic consumption has often been problematic in international trade literature. I overcome this difficulty by complementing the bilateral trade dataset with the FAO food balance sheet.¹⁰ The FAO food balance sheet provides information on total production, domestic consumption, total exports, and total imports by type of commodities after 2010. I merge the food balance sheet data with bilateral trade flows and infer the domestic consumption so that the trade patterns are consistent in the two datasets.

⁹For instance, I observe how much tuna was exported from China to USA from the FAO dataset on fishery trade.

¹⁰The purpose of FAO food balance sheet is to understand the current status of food consumption and security by each country.

Other Fishery Datasets. Other fishery datasets used in the paper include fishery habitat suitability, fishery ex-vessel prices, fishery subsidies, and fishery employment. The fishery habitat suitability, provided by Sea Around Us, measures the time-invariant habitat suitability of overall geographic conditions at the grid for over 1,000 species.¹¹ The fishery ex-vessel prices, also published by Sea Around Us, measure the landed value of fishery at the port for over 1,000 species, expressed in nominal USD per tonne since 1980.¹² While the fishery ex-vessel prices do not vary across fishing locations, they provide a consistent comparison of landed value across countries and species (Sumaila et al., 2015). The country-level fishery subsidies data, sourced from Sumaila et al. (2019), are available from 2002 to 2018. Lastly, fishery employment data are from the OECD since 2010.

Geographic Boundaries. I use the geographic boundary shapefiles to assign $0.5^\circ \times 0.5^\circ$ grids into EEZs and high seas. To assign the grids into EEZs, I use the shapefile of EEZ boundaries from the version 12 of Flanders Marine Institute.¹³ To assign the grids into high seas, I use the shapefile from FAO major fishing area boundaries.¹⁴

Other Country-level Datasets. I use data on GDP and total employment from World Economic Outlook and UN-ILO, respectively. I also obtain the coordinates of global ports from World Port Index, covering more than 3,500 major ports worldwide.

Harmonization of Datasets. For the rest of the paper, I harmonize the dataset in the following way. First, I classify countries into 30 countries, consisting of 25 individual countries and 5 regional aggregates. 25 individual countries are picked based on the national GDP and fishery output, covering 78% of global GDP and 84% of global fishery output. The remaining countries are regrouped into 5 regional aggregates based on their continents. Appendix Table A.1 provides the mapping of countries.

Second, I classify $0.5^\circ \times 0.5^\circ$ grids of the global ocean into 46 fishing locations, consisting of 30 EEZs and 16 high seas. Each EEZ maps to the EEZ of each country, picked earlier based on the size of the overall economy and fishery industry. Each high sea corresponds to each of FAO major fishing areas that are exclusive of EEZs. By overlaying the shapefile of EEZs with the shapefile of FAO major fishing areas, I define each high sea as the grids

¹¹The geographic conditions include but are not limited to depth, distance from the coast, and the existence of coral reefs (Zeller and Pauly, 2015).

¹²The dataset is constructed by compiling various data sources including FAO, OECD, the European Commission, etc.

¹³The shapefile of EEZ boundaries can be downloaded at <https://www.vliz.be/en/imis?dasid=8394&doiid=911>.

¹⁴The shapefile of FAO major fishing area boundaries can be downloaded at <https://www.fao.org/fishery/en/area/search>

belonging to each of FAO major fishing areas but not to any of EEZs. Appendix Table A.2 provides the list of 46 fishing locations.

Lastly, I aggregate individual species into 10 species. While the fishery catch dataset covers more than 1,100 species and the fishery trade dataset covers about 1,000 products, I harmonize the dataset by mapping them into 10 species from the International Standard Statistical Classification of Aquatic Animals and Plants (ISSCAAP). Appendix Table A.3 provides the list of 10 species. As a result, the harmonized dataset consists of 30 countries, 46 fishing locations, and 10 species.

3 Empirical Patterns

In this section, I document three empirical patterns of global fishery that motivate the quantitative model. First, I show the decline in fishery stock between 1980 and 2020. Second, I find that the lack of property rights is associated with lower fishery stock. Third, I document that fishery subsidies are correlated with fishing on high seas.

Pattern 1. Global fishery stock has been decreasing

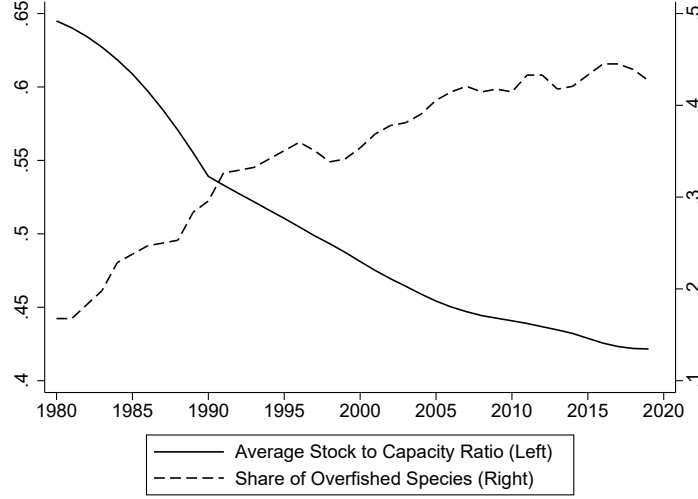
Figure 2 shows the trend in global fishery stock between 1980 and 2018. The solid line refers to the average stock to capacity ratio across species and fishing locations. The stock to capacity ratio measures a stock level of fishery relative to its carrying capacity for a given species and fishing location. This ratio serves as an indicator of anthropogenic impact on fishery stock since it approaches 1 in the absence of harvest and approaches 0 as the stock nears complete depletion. During this period, the average stock to capacity ratio decreased from 0.65 to 0.42, implying a 35% decrease on average. The dotted line refers to the proportion of overfished species. Following the criteria from FAO, the species are defined to be overfished if the stock level falls below 40% of carrying capacity. I find that the proportion of overfished species increased from 17% to 43% during this period. Appendix Table B.4 provides detailed descriptive statistics of fishery stock changes by species.¹⁵

Pattern 2. Assigning property rights lead to larger fishery stock

Next, I examine the relationship between property rights and fishery stock. To do so, I employ the following event-study framework, taking the declaration of EEZ in 1994 as a

¹⁵Appendix Table B.4 suggests the heterogeneity in stock changes across species. Some stocks, such as squids, experienced more than 40% decrease in stock, while some stocks, such as cods, only decreased by 6%.

Figure 2: Global Fishery Stock over Time (1980-2018)



Notes: Species and fishing locations are aggregated into 10 species and 46 fishing locations. A species by region is defined to be overfished if the stock is below 40% of carrying capacity, following the criteria from FAO (FAO, 2022).

quasi-experiment. The declaration of EEZ in 1994 provided coastal nations with property rights over marine resources within 200 nautical miles of their shores.¹⁶ I argue that the declaration of EEZ was exogenous to the fishery stock, as the focus of the introduction of EEZ was primarily on energy and mineral resources due to the relative size of sectors (Osherenko, 2006).

Taking the declaration of EEZ in 1994 as a quasi-experiment, I estimate equation (1) to study the effect of property rights on fishery stock

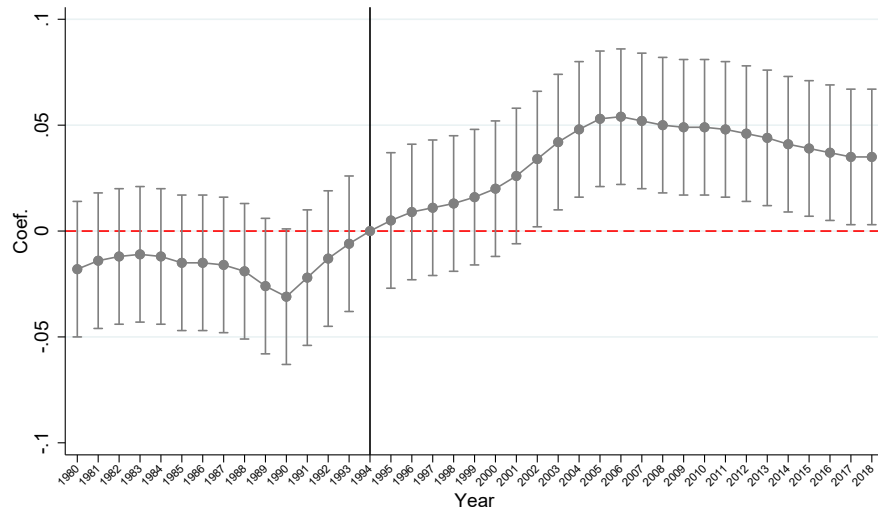
$$x_{k,st}/M_{k,s} = \beta_0 + \sum_{\tau=1980}^{2018} \beta_{\tau} EEZ_k \times \mathbb{1}\{t = \tau\} + \delta_{k,s} + \delta_t + \varepsilon_{k,st} \quad (1)$$

where $x_{k,st}/M_{k,s}$ is the stock to capacity ratio at fishing location k for species s in year t , EEZ_k is the indicator variable for whether fishing location k belongs within the boundaries of EEZ, $\mathbb{1}\{t = \tau\}$ is the indicator variable for whether year $t = \tau$, $\delta_{k,s}$ is the fishing location-species fixed effect, δ_t is the year fixed effect, and $\varepsilon_{k,st}$ is the error term. Using 1994 as a base year, the coefficient β_{τ} measures the difference in the average stock to capacity ratio at the EEZ relative to the high sea in year τ compared to the base year. Given that the stocks should not have been disproportionately affected before the declaration of EEZ, I expect the coefficient β_{τ} to be zero for $\tau < 1994$.

Figure 3 presents the estimates of β_{τ} from equation (1). Prior to the EEZ declaration

¹⁶While EEZs were first defined in the 1982 United Nations Convention on the Law of the Sea, they were formally declared and became effective in 1994.

Figure 3: Declaration of EEZ and Fishery Stock



Notes: The figure shows the coefficients for the event study regression of equation (1). Each dot indicates the point estimate of β_τ and each range refers to the 95% confidence interval, where β_τ captures the difference in the average stock to capacity ratio at the EEZ relative to the high sea in year τ compared to the year of EEZ declaration.

in 1994, the coefficient β_τ is not statistically different from zero, validating the absence of differential trends in fishery stocks between the EEZ and high sea. The coefficient β_τ becomes significantly positive after 2002, indicating that the EEZ experienced a differential increase in fishery stocks. The delayed emergence of significant effects suggests a temporal lag in the realization of fishery stock increase. Quantitatively, in the years after 2002, the stock to capacity ratio in the EEZ increased by approximately 5 percentage points relative to the high sea, compared to the base year 1994. Given that the average stock to capacity ratio in 1994 was 0.52 from Figure 2, 5 percentage points increase account for about 10% increase in average fishery stock.

This pattern motivates a model in which a lack of property rights leads to exploiting fishery stock. Thus, I compare two polar cases of property rights in the model: the decentralized equilibrium, where firms are atomistic with open access, and the socially optimal allocation, where the planner has globally exclusive property rights.

Pattern 3. Fishery subsidies are correlated with high sea production

Finally, I investigate the impact of fishery subsidies on the location of fishing. Despite global efforts to preserve fishery stocks (UN, 1982, 2023), individual countries continue to promote fishing activities, mainly through subsidies. While the first discussion to regulate fishery subsidies began in the 2001 Doha Round (Chang, 2003), effective global regulation has not been achieved. Consequently, the fishery remains highly subsidized, with subsidies

Table 1: Fishery Subsidies and High Sea Production (2002-2018)

	(1)	(2)	(3)	(4)
$\Delta \ln S_{i,t}$	0.346*	0.323*	0.117**	0.112**
	(0.158)	(0.163)	(0.040)	(0.042)
Other Controls	N	Y	N	Y
Type of Subsidies	All	All	Fuel	Fuel
R^2	0.132	0.176	0.187	0.268
N	30	30	30	30

Notes: The regression is weighted by log output in the initial year 2002. Columns “All” refer to the regression where the regressor includes all fishery subsidies. Columns “Fuel” refer to the regression where the regressor includes fishery subsidies. Other controls include log differences of population and GDP per capita.

accounting for 10% of global fishery output (Sala et al., 2018).¹⁷

Among the major types of fishery subsidies, fuel subsidies are important for the following reasons.¹⁸ First, they constitute the largest single category of subsidies, representing 21% of total subsidies in the fishing industry in 2018. Second, they have become the primary target for global regulation due to their role in facilitating the exploitation of resources in the high sea by extending the operational range of fishing fleets (Sumaila et al., 2019). Appendix Table B.5 shows the summary statistics of fuel subsidies between 2002 and 2018.¹⁹

I consider equation (2) to explore the relationship between subsidies and the location of fishing:

$$\Delta \ln(Y_{i,t}^{HS}/Y_{i,t}^{EEZ}) = \beta_0 + \beta_1 \Delta \ln S_{i,t} + \Delta X_{i,t} + \varepsilon_{i,t} \quad (2)$$

where $Y_{i,t}^{HS}$ and $Y_{i,t}^{EEZ}$ are the high sea and EEZ production for country i at year t , $S_{i,t}$ is the fishery subsidy, $X_{i,t}$ is other controls, and $\varepsilon_{i,t}$ denotes the error term. The operator Δ refers to the change between 2002 and 2018. The coefficient β_1 captures the correlation between fishery subsidies and the relative production at the high sea.

There could be a concern about the confounding factors that affect the incentives to provide subsidies and the fishing locations. For example, if EEZs were fully exploited due to higher demand for fishery, governments could have larger incentives to provide subsidies, affecting the amount of subsidies and the relative production at the high sea. To address the endogeneity issue, I control for population growth and GDP growth.

Table 1 shows the estimates of equation (2) for different types of subsidies. Columns

¹⁷Following the continued concern, the WTO has recently made another agreement to prohibit subsidies that promote fishing (WTO, 2023).

¹⁸The major types of subsidies include fuel, infrastructure, and management subsidies (Sumaila et al., 2019).

¹⁹China has more than doubled its fuel subsidies during this period, accounting for more than 40% of total fuel subsidies in 2018.

(1) and (2) include all types of fishery subsidies as the regressor, while Columns (3) and (4) restrict the regressor to fuel subsidies. The results suggest that countries that increase their fishery subsidies tend to disproportionately expand their fishery output in high seas. After controlling for population growth and GDP growth, column (2) suggests that a 10 percent increase in fishery subsidies is associated with a 3 percent increase in the output from high sea relative to EEZ. Column (4) implies that a 10 percent increase in fuel subsidies is correlated with an 1.1 percent increase in the output from high sea relative to EEZ.

Motivated by this empirical fact, I examine the impact of fuel subsidies on global welfare and the spatial distribution of fishing activities in the quantitative exercise. Moreover, the regression is informative about the elasticity of substitution across fishing locations. The idea is that if the fishery goods are more substitutable across locations, the effect of subsidies on relative production would be larger. In Section 5, I simulate the model and estimate equation (2) using the simulated results to discipline the parameter governing the elasticity of substitution across fishing locations.

4 A Model of Global Fishery

In this section, I develop a dynamic spatial model of global fishery, where each country produces from fishing locations and exports to another country. I compare two polar cases: a decentralized equilibrium, where atomistic firms have open access to the fishery, and the socially optimal allocation, where a social planner has exclusive property rights. While the atomistic firms solve a static problem, the global social planner solves a dynamic problem, fully internalizing the social cost.²⁰ I describe the environment and characterize the decentralized equilibrium and the planner's problem.

4.1 Environment

The economy consists of I countries indexed by i or j . There are H high sea locations, which are the sea locations that are beyond any country's jurisdiction. Since I and H are disjoint, there are in total $K = I + H$ fishing locations, indexed by k . Time is discrete and indexed by t . The economy has two sectors: fishery, f , and outside good, o . The fishery sector is disaggregated into $s \in S$ species. Outside good is freely traded and serves as numeraire. The frictions in the economy are iceberg commuting costs $\tau_{ik,s}$ and iceberg trade costs $d_{ij,s}$. The iceberg commuting costs $\tau_{ik,s}$ are the frictions that exporters face upon harvesting at fishing

²⁰The status quo can be interpreted as in-between two polar cases, where individual countries internalize some dynamic costs within their EEZs.

locations and bringing back the output to their ports. The iceberg trade costs $d_{ij,s}$ are the frictions that exporters face upon shipping fishery to importers. The economy is endowed with the initial fishery stock $x_{k,s0}$, distinguished by fishing location k and species s , and the total labor force $\tilde{L}_i$. For expositional purposes, I suppress the subscript f for the fishery sector.

4.1.1 Preferences

The representative household in importer j has quasi-linear preferences between outside good and fishery bundle:

$$U_{j,t} = C_{j,t}^o + b_j \ln C_{j,t} \quad (3)$$

where $U_{j,t}$ is the household welfare at time t , $C_{j,t}^o$ is the consumption of outside good, $C_{j,t}$ is the consumption of fishery bundle, and b_j is the demand shifter of the fishery bundle. One advantage of quasi-linear preferences is that the equilibrium at the fishery sector is independent of income effects.

The fishery bundle is comprised of three tiers of CES aggregators. In the upper tier, varieties of fishery goods are differentiated by species s , for example, tuna and squid. Consumers in importer j combine the varieties from every species s according to CES preferences with the elasticity of substitution ν and the demand shifters b_s :

$$C_{j,t} = \left(\sum_{s \in S} b_s C_{j,st}^{\frac{\nu-1}{\nu}} \right)^{\frac{\nu}{\nu-1}} \quad (4)$$

where $C_{j,st}$ is the consumption of fishery species s .

In the middle tier, varieties of species s are differentiated by exporter i , for example, tuna exported by China and tuna exported by Chile. Consumers in country j consuming species s combine the varieties from every exporter i according to CES preferences with the elasticity of substitution η and the demand shifters b_{ij} :

$$C_{j,st} = \left(\sum_{i \in I} b_{ij} C_{ij,st}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \quad (5)$$

where $C_{ij,st}$ is the consumption of fishery species s from exporter i .

Finally, in the lower tier, varieties of species s from exporter i are differentiated by fishing location k , for example, tuna exported by China caught from Pacific ocean and tuna exported by China caught from Atlantic ocean. Consumers in importer j consuming species s by exporter i combine the varieties from every fishing location k according to CES preferences

with the elasticity of substitution κ and the demand shifters $b_{k,s}$:

$$C_{ij,st} = \left(\sum_{k \in K} b_{k,s} C_{ijk,st}^{\frac{\kappa-1}{\kappa}} \right)^{\frac{\kappa}{\kappa-1}} \quad (6)$$

where $C_{ijk,st}$ is the consumption of fishery species s from exporter i at fishing location k .

4.1.2 Technology

In the fishery sector, each variety of fish is produced under the constant to returns to scale technology:

$$Q_{ijk,st} = \frac{z_i (x_{k,st-1}/M_{k,s})^\xi}{d_{ij,s} \tau_{ik,s}} L_{ijk,st} \quad (7)$$

where $Q_{ijk,st}$ is the output quantity of fishery by exporter i to importer j at fishing location k for species s , $x_{k,st-1}$ is the fishery stock from previous period, $M_{k,s}$ is the carrying capacity, $d_{ij,s}$ is the iceberg trade cost, $\tau_{ik,s}$ is the iceberg commuting cost, z_i is the country-specific productivity shifter, ξ is the stock elasticity of output, and $L_{ijk,st}$ is the labor input.

Since the output $Q_{ijk,st}$ represents the quantity delivered to the importer, the fishery technology is subject to the iceberg trade and commuting costs. Specifically, delivering one unit of fish to importer j requires exporter i to ship $d_{ij,s}$ units of fish under the iceberg trade cost. Additionally, shipping $d_{ij,s}$ units of fish requires exporter i to harvest $d_{ij,s} \tau_{ik,s}$ units of fish from the fishing location k under the iceberg commuting cost.

Moreover, the fishery productivity is an increasing function of the stock to capacity ratio $x_{k,st-1}/M_{k,s}$. This captures the efficiency gains in the harvest from stock abundance with an elasticity of ξ .

In the outside good sector, outside good is produced under the constant to returns to scale technology:

$$Q_{i,t}^o = z_i^o L_{i,t}^o \quad (8)$$

where z_i^o is the productivity shifter, and $L_{i,t}^o$ is the labor input. I assume that the outside sector is large enough to be produced at every country.

4.1.3 Nature

Following the standard literature on population dynamics ([Gordon, 1954](#); [Schaefer, 1954](#); [Clark, 1990](#); [Smith, 2009](#)), the evolution of fishery stock is determined by two competing

forces: ecological growth function and anthropogenic activity such that

$$x_{k,st+1} - x_{k,st} = G(x_{k,st}) - H(x_{k,st}) \quad (9)$$

where $x_{k,st}$ is the fishery stock, $G(x_{k,st})$ is the natural regrowth, and $H(x_{k,st})$ is the total harvest.

The natural regrowth $G(x_{k,st})$ follows a logistic equation capturing two fundamental biological mechanisms in fishery dynamics:

$$G(x_{k,st}) = \gamma_{k,s} x_{k,st} \left(1 - \frac{x_{k,st}}{M_{k,s}}\right) \quad (10)$$

where $\gamma_{k,s}$ is the intrinsic growth rate and $M_{k,s}$ is the carrying capacity. First, at low stock levels, population grows proportionally at the intrinsic growth rate $\gamma_{k,s}$, reflecting the reproductive ability of species. Second, at high stock levels, population growth is bounded by the carrying capacity $M_{k,s}$, representing environmental constraints to species.²¹

The total harvest $H(x_{k,st})$ equals the sum of harvests across all bilateral trade pairs adjusted for trade and commuting frictions:

$$H(x_{k,st}) = \sum_{i \in I} \sum_{j \in I} d_{ij,s} \tau_{ik,s} Q_{ijk,st} \quad (11)$$

where $d_{ij,s}$ is the iceberg trade cost, $\tau_{ik,s}$ is the iceberg commuting cost, and $Q_{ijk,st}$ is the output quantity. The total harvest is obtained by multiplying the output by the iceberg trade and commuting frictions and summing over all importers and exporters, since the output $Q_{ijk,st}$ represents the quantity delivered to importer j .

4.2 Decentralized Equilibrium

In this subsection, I characterize the decentralized equilibrium where fishery firms are atomistic under open access to the fishery. All markets are perfectly competitive. The outside good is freely traded and serves as numeraire. Each atomistic firm solves a static problem, ignoring the impact of its harvest on future stock. The only dynamic factor is the evolution of fishery stock in the decentralized equilibrium. The details are provided in Appendix The details are provided in Appendix C.1.

²¹Given these features, the stock to capacity ratio $x_{k,st}/M_{k,s}$ ranges from 0 to 1. The first partial derivative of fishery growth is a decreasing function of $x_{k,st}$ and the second partial derivative is negative, with the fishery growth being maximized at $x_{k,st}/M_{k,s} = 0.5$.

4.2.1 Utility Maximization

Given equations (3), (4), (5), and (6), the representative household in country j maximizing its utility allocates a fixed amount of its expenditure b_j on fishery bundle and the rest to the outside good such that

$$X_{j,t} = b_j, \quad X_{j,t}^o = E_{j,t} - b_j$$

where $X_{j,t}$ is the fishery expenditure and $E_{j,t}$ is the household income.

From the upper tier of CES, the expenditure share of species s in all fishery expenditure by country j , $\pi_{j,st}$, and the consumer price index in country j for aggregate fishery bundle, $P_{j,t}$, are given by

$$\pi_{j,st} = \frac{b_s P_{j,st}^{1-\nu}}{P_{j,t}^{1-\nu}}, \quad P_{j,t} = \left[\sum_{s \in S} b_s P_{j,st}^{1-\nu} \right]^{\frac{1}{1-\nu}}$$

Then, from the middle tier of CES, the expenditure share of country i in all expenditure by country j for species s , $\pi_{ij,st}$, and the consumer price index in country j for species s , $P_{j,st}$, are given by

$$\pi_{ij,st} = \frac{b_{ij} P_{ij,st}^{1-\eta}}{P_{j,st}^{1-\eta}}, \quad P_{j,st} = \left[\sum_{i \in I} b_{ij} P_{ij,st}^{1-\eta} \right]^{\frac{1}{1-\eta}}$$

Finally, from the bottom tier of CES, the expenditure share of fishing location k in all expenditure by country j for species s from country i , $\pi_{ijk,st}$, and the consumer price index in country j for species s from country i , $P_{ij,st}$, are given by

$$\pi_{ijk,st} = \frac{b_{k,s} p_{ijk,st}^{1-\kappa}}{P_{ij,st}^{1-\kappa}}, \quad P_{ij,st} = \left[\sum_{k \in K} b_{k,s} p_{ijk,st}^{1-\kappa} \right]^{\frac{1}{1-\kappa}}$$

where $p_{ijk,st}$ equals the marginal cost of production for species s from exporter i to importer j at fishing location k .

4.2.2 Profit Maximization

Given equation (7), for each variety of fish, a representative firm in the fishery sector chooses its labor inputs, thus its output, to maximize the profit. I assume that each firm is atomistic under open access. Instead of solving a dynamic problem, each atomistic firm solves a static problem, ignoring the impact of its harvest on future stock. The price of fishery $p_{ijk,st}$ equals the marginal cost of production such that

$$p_{ijk,st} = \frac{d_{ij,s} \tau_{ik,s}}{(x_{k,st-1}/M_{k,s})^\xi} \frac{w_{i,t}}{z_i} \quad (12)$$

where $w_{i,t}$ is the wage of fishery worker in exporter i .

Given equation (8), a representative firm in the outside good sector chooses its labor inputs, thus its output, to maximize the profit. The outside good is freely traded and serves as numeraire. I assume that the outside good sector is large enough and the fishery sector is relatively small such that the outside good is produced in all countries.

4.2.3 Labor Mobility

I assume a perfect mobility of workers across sectors. Since the outside good serves as numeraire, the wage that applies to all workers is exogenously determined as the productivity shifter of outside good such that

$$w_i^o = w_i = z_i^o \quad (13)$$

where w_i is the wage of outside good worker in exporter i .

4.2.4 Market Clearing

Goods market clearing on fishery sector requires that the revenue equals the expenditure for each variety of fishery:

$$Y_{ijk,st} = \pi_{ijk,st} \pi_{ij,st} \pi_{j,st} X_{j,t} \quad (14)$$

where $Y_{ijk,st}$ is the fishery revenue and $X_{j,t}$ is the total expenditure on fishery in importer j

Goods market clearing on outside good sector requires that the sum of revenue across all exporters equal the sum of expenditure across all importers:

$$\sum_{i \in I} Y_{i,t}^o = \sum_{j \in I} X_{j,t}^o \quad (15)$$

where $Y_{i,t}^o$ is the outside good revenue in exporter i and $X_{j,t}^o$ is the outside good expenditure in importer j .

Labor market clearing requires that payments to labor equal revenue for each sector:

$$w_{i,t} L_{ijk,st} = Y_{ijk,st}, \quad w_{i,t}^o L_{i,t}^o = Y_{i,t}^o \quad (16)$$

Lastly, the balance of household budget holds that the sum of factor rewards equals the sum of expenditures:

$$\sum_{j \in I} \sum_{s \in S} \sum_{k \in K} w_{i,t} L_{ijk,st} + w_{i,t}^o L_{i,t}^o = X_{i,t} + X_{i,t}^o \quad (17)$$

4.2.5 Equilibrium

Definition. [Competitive equilibrium] Given taste and geography, initial vector of fish stock $\{x_{k,s0}\}$ and labor endowment $\{\bar{L}_i\}$, a **competitive equilibrium** is a path of consumption $\{C_{ijk,st}, C_{i,t}^o\}$, output $\{Q_{ijk,st}, Q_{i,t}^o\}$, labor allocation $\{L_{ijk,st}, L_{i,t}^o\}$, prices $\{p_{ijk,st}\}$, wages $\{w_{i,t}, w_{i,t}^o\}$, and fish stock $\{x_{k,st}\}$, such that household maximizes its utility according to (3), firms maximize their profits according to (7) and (8), stock of fish evolves according to (9), markets clear according to (14) and (15) and (16), and the balance of budget holds according to (17)

Definition. [Steady-state equilibrium] A **steady state equilibrium** is the allocation that satisfies the conditions of competitive equilibrium in addition to the condition where the regrowth of fishery equals with total harvest such that

$$\gamma_{k,s} x_{k,st} \left(1 - \frac{x_{k,st}}{M_{k,s}}\right) = \sum_{i \in I} \sum_{j \in I} d_{ij,s} \tau_{ik,s} Q_{ijk,st} \quad \forall k, s \quad (18)$$

4.2.6 Fishery Consumption in the Decentralized Equilibrium

Rearranging the first-order conditions from households and firms, I can characterize the fishery consumption in the decentralized equilibrium as:

$$\underbrace{\frac{\partial U_{j,t}}{\partial C_{ijk,st}}}_{\text{MU of fishery consumption}} = \underbrace{\frac{\partial U_{j,t}}{\partial C_{j,t}^o} \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} / \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right)}_{\text{MC of fishery consumption}} \quad (19)$$

Equation (19) implies that the fishery consumption for each variety is determined where the marginal utility equals the marginal cost of fishery consumption. Importantly, the decentralized equilibrium fails to internalize the impact of harvest today on future stock. As a result, the marginal cost of fishery consumption is purely static, representing the static opportunity cost of inputs that could have been used to produce outside good if the fishery were not produced. This expression will be useful when comparing the outcomes with the planner allocation in Section 4.3, where the planner internalizes the dynamic impact of harvest.

4.3 Planner's Problem

This subsection characterizes the planner's problem, where the planner who owns exclusive property rights over global fishery stock maximizes the discounted present value of global

welfare. I analytically provide the expression comparing the outcomes from the socially optimal allocation with the decentralized equilibrium.

4.3.1 Planner's FOC

Given the GE equations (3)~(11), define the global welfare W_t as

$$W_t = \sum_{i \in I} \phi_i U_{i,t} \quad (20)$$

where $\phi_i \geq 0$ is the Pareto weights. ²²

The planner allocates the labor, consumption, and fishery stock to maximize the net present value of global welfare such that

$$\begin{aligned} \max_{C_{ijk,st}, C_{j,t}^o, L_{ijk,st}, L_{i,t}^o, x_{k,st}} \sum_t \beta^t W_t \quad \text{s.t.} \quad & [\lambda_{k,st}] \quad x_{k,st-1} + G_{k,st} = H_{k,st} + x_{k,st} \\ & [\mu_{i,t}] \quad \bar{L}_i = \sum_{j \in I} \sum_{s \in S} \sum_{k \in K} L_{ijk,st} + L_{i,t}^o \\ & [\theta_{ijk,st}] \quad Q_{ijk,st} = C_{ijk,st} \\ & [\alpha_t] \quad \sum_i Q_{i,t}^o = \sum_j C_{j,t}^o \end{aligned} \quad (21)$$

where $\lambda_{k,st}, \mu_{i,t}, \theta_{ijk,st}, \alpha_t$ denote the Lagrangian multipliers for stock constraint, labor constraint, and technology constraint for fishery and outside good, respectively.

4.3.2 Fishery Consumption in the Planner Allocation

In this subsection, I compare the first-order conditions of the planner allocation and the decentralized equilibrium. Provided that the interior solution exists, rearranging the first-order conditions of the planner at the steady state yields:

$$\underbrace{\frac{\partial U_{j,t}}{\partial C_{ijk,st}}}_{\text{MU of fishery consumption}} = \underbrace{\frac{\partial U_{j,t}}{\partial C_{j,t}^o} \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} / \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right)}_{\text{static MC}} + \underbrace{\frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}}}_{\text{dynamic MC}} \quad (22)$$

MC of fishery consumption

where $\Omega_{k,st} := \partial W_{t+1} / \partial x_{k,st}$ and $\Theta_{k,st} := \partial x_{k,st+1} / \partial x_{k,st}$. The allocation of outside good is undetermined with the symmetric Pareto weights where $\phi_i = 1$. The derivation details are

²²Without loss of generality, I assume that $\phi_i = 1$ is symmetric across countries to focus on the interior solution. The implication of symmetric ϕ_i is that the allocation of outside good is undetermined under the planner allocation. When ϕ_i is asymmetric, all of the outside good is allocated to the country with the largest ϕ_i , generating the corner solution.

provided in Appendix C.2.

For each variety of fishery $C_{ijk,st}$, the planner allocates the fishery consumption until the marginal utility equalizes with the marginal cost of the fishery consumption. The marginal cost of the fishery consumption consists of two terms: the static marginal cost and the dynamic marginal cost.

In equation (22), the first term on the right-hand side represents the static marginal cost, which is the opportunity cost of inputs that could have been used to produce the outside good instead of the fishery. While both the planner and decentralized equilibrium account for this static cost, the key difference lies in the second term: the dynamic marginal cost. This term captures the welfare loss from reduced future productivity as harvest today depletes fishery stocks.

The planner internalizes the dynamic marginal cost by explicitly accounting for the effect of harvest today on future fishery stocks. In contrast, the decentralized equilibrium disregards these future impacts, failing to internalize the dynamic marginal cost. The market failure to internalize the dynamic marginal cost becomes the source of externalities in global fisheries. Given that this dynamic marginal cost is positive and the fishery consumption exhibits diminishing marginal utility, the planner allocates fewer inputs to the fishery and maintains larger fishery stocks compared to the decentralized equilibrium.

The size of the dynamic marginal cost at the steady state depends on three terms: (i) the welfare loss per unit of stock decrease and (ii) the amount of stock decrease given one unit of fishery consumption.

First, the welfare loss per unit of stock decrease, $\Omega_{k,st} = \partial W_{t+1} / \partial x_{k,st}$, captures how much of the welfare decreases through the lower productivity given one unit decrease in stock. Given the fishery technology in equation (7), the fishery sector becomes less productive as the fishery stock decreases and the stock elasticity of output ξ governs the size of the effect. The lower productivity decreases the welfare from all countries in the economy. If there were no productivity gains from fishery stock ($\xi = 0$), there would be no externalities, implying that the planner allocation would coincide with the decentralized equilibrium.

Second, the total size of welfare loss depends also on the amount of stock decrease given one unit of fishery consumption. Given equation (11), one unit of fishery consumption requires to decrease the stock by $\partial H_{k,st} / \partial Q_{ijk,st} = d_{ij,s} \tau_{ik,s}$ due to the iceberg trade and commuting frictions. Since regrowth and harvest of fish depend on the stock level from equation (9), the change in stock at the next period, given one unit change of stock today, is not necessarily one. However, it is endogenously determined by $\Theta_{k,st} = \partial x_{k,st+1} / \partial x_{k,st}$, which captures the change in the number of stocks next period, given one unit change in fishery stock today. Given a unit decrease in fishery stock today, the number of stock in the

next period decreases by $\Theta_{k,st}$, and the effect is accumulated over periods. At the steady state, the total number of affected stock becomes $1/(1 - \beta\Theta_{k,st})$ units in the formulation of infinite sum up to the discount rate.

Finally, I point out the sufficient condition for $\Theta_{k,st}$ to converge.

Lemma. *At the steady state, sufficient conditions for $\theta_{k,st}$ to converge are i) $\xi \geq 1$ or ii) $0 < \xi < 1$ and $\gamma_{k,s} < \frac{1}{1-\xi} \left(\frac{1}{\beta} - 1 \right)$*

Proof. See Appendix C.4. □

The intuition is as follows. When ξ is large enough, given a unit increase in stock, the harvest increases enough to suppress $\Theta_{k,st}$. When ξ is small, since the harvest does not suppress $\Theta_{k,st}$ as much as in the previous case, smaller $\gamma_{k,s}$ is required so that the regrowth does not respond too rapidly given a unit increase in stock. As ξ approaches zero, the required $\gamma_{k,s}$ decreases and gets closer to $1/\beta - 1$.

4.4 Optimal Policy

The First and Second Welfare Theorem guarantee that the following two approaches yield identical outcomes: the primal approach, where the planner directly chooses optimal allocation, and the dual approach, where the planner sets the optimal tax and the lump-sum international transfers (Ljungqvist and Sargent, 2018). In this subsection, I extend the decentralized equilibrium with the tax and derive the expression for the optimal policy.

The tax $t_{ijk,st}$ additively increases the marginal cost of production so that the price after the tax $\tilde{p}_{ijk,st}$ becomes

$$\tilde{p}_{ijk,st} = p_{ijk,st} + t_{ijk,st} \quad (23)$$

where $p_{ijk,st}$ is the price from the decentralized equilibrium without the tax as described in equation (12). Since the tax $t_{ijk,st}$ is specific to the destination, this tax could be interpreted as the combination of domestic production tax and export tax.

The collected taxes are redistributed to the households as a lump-sum transfer. In turn, the balance of budget condition for households in equation (17) now becomes

$$\sum_{j \in I} \sum_{s \in S} \sum_{k \in K} w_{i,t} L_{ijk,st} + w_{i,t}^o L_{i,t}^o + \sum_j \sum_s \sum_k t_{ijk,st} Q_{ijk,st} = X_{i,t} + X_{i,t}^o \quad (24)$$

Following the dual approach, the planner can choose the optimal tax $t_{ijk,st}^*$ that achieves the socially optimal allocation from Section 4.3.1. There is no need to impose the lump-sum

international transfers. Under the quasi-linear preferences, there is no income effect in the fishery consumption, and the lump-sum international transfer only affects the consumption of outside good across countries. However, in the planner allocation with the symmetric Pareto weights $\phi_i = 1$, the allocation of outside good is undetermined since the planner is indifferent to the consumption of outside good across countries. Therefore, any set of international transfers, including zero transfers across all countries, is consistent with the socially optimal allocation.²³

Respecting the first-order conditions from the decentralized equilibrium, the optimal tax $t_{ijk,st}^*$ equals the dynamic marginal cost from equation (22).

$$t_{ijk,st}^* = \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}}$$

where the details are explained in Appendix C.3.

5 Taking the Model to Data

This section provides the calibration of the quantitative model. As described in Section 2, I aggregate the world into 30 countries, the global oceans into 46 fishing locations, and the fishery species into 10 species. I take the model to data in 2018 to calibrate the model. I begin by describing the calibration of elasticities of substitutions, and proceed to the calibration of remaining parameters.

5.1 Calibration of Elasticities of Substitutions

In this subsection, I calibrate the parameters governing the elasticities of substitutions. In the spirit of Costinot et al. (2016), I use a supply-side instrument for productivity to identify the demand relationship in the gravity equation. I start by estimating the elasticity of substitution across countries η , then estimate the elasticity of substitution across species ν . Lastly, I estimate the elasticity of substitution across fishing locations κ using the indirect inference disciplined by equation (2) about the relationship between the subsidies and the location of fishing.

²³If the Pareto weights were not symmetric, the lump-sum international transfer is necessary so that the outcome is consistent with the corner solution by the planner where a country with the largest weight consumes all of the outside good.

5.1.1 Elasticity of Substitution across Exporters (η)

I estimate the elasticity of substitution across exporters η from the following gravity equation.

$$\log \left(\frac{X_{ij,s}}{X_{j,s}} \right) = (1 - \eta) \log p_{i,s} + \phi_{j,s} + \phi_{ij} + \varepsilon_{ij,s} \quad (25)$$

where $X_{ij,s}/X_{j,s}$ is the expenditure share for exporter i in all expenditure by importer j for species s , $p_{i,s}$ is the producer prices at the border of exporter i for species s , $\phi_{j,s}$ is the importer-species fixed effect, ϕ_{ij} is the exporter-importer fixed effect, and $\varepsilon_{ij,s}$ is the residual term. I observe the expenditure share $X_{ij,s}/X_{j,s}$ from the trade dataset, and the producer prices at the border $p_{i,s}$ from the ex-vessel prices dataset, which is invariant cross fishing locations. In particular, the producer prices at the border $p_{i,s}$ is:

$$p_{i,s} = \left[\sum_{k \in K} b_{k,s} (p_{ik,s})^{1-\kappa} \right]^{\frac{1}{1-\kappa}} \quad (26)$$

where $p_{ik,s}$ equals to the marginal cost of production by exporter i at fishing location k for species s . From the exporter-importer fixed effect $\phi_{j,s}$ and the residual term $\varepsilon_{ij,s}$, I recover the exporter-importer specific preference shifter b_{ij} and the trade cost $d_{ij,s}$ such that $b_{ij} = \exp(\phi_{ij})$ and $d_{ij,s} = \exp\left(\frac{1}{1-\eta}\varepsilon_{ij,s}\right)$.

A concern about endogeneity arises because prices can be correlated with unobserved demand shifters. To mitigate this concern, I adopt a supply-side IV, which is correlated with the producer prices at the border but uncorrelated with an unobserved demand shifter. Specifically, I instrument the producer prices at the border $p_{i,s}$ with the proximity to suitable habitats $Z_{i,s}$ by exporter i for species s .

I construct the instrument, the proximity to suitable habitats $Z_{i,s}$, using the grid level data on habitat suitability and the linear distance between each grid and the closest port of countries. The grid level data on habitat suitability $Z_{g,s}$ measures how suitable habitat a grid g is for species s , using the geographic conditions including ocean depth, temperature, and climate. I construct the country level measure for proximity to habitat suitability $Z_{i,s}$:

$$Z_{i,s} = \sum_{g \in G} \frac{Z_{g,s}}{\log dist_{ig}}$$

where $dist_{ig}$ is the linear distance to each grid g from the closest port at country i . To construct the country level measure for proximity to habitat suitability, I take the weighted average of the habitat suitability $Z_{g,s}$ across grids, where the inverse of weights is the log of $dist_{ig}$.

Table 2: Estimation of η

	(1) PPML	(2) OLS	(3) IV	(4) IV	(5) IV
$\log P_{is}$	-0.386*** (0.057)	-0.808*** (0.078)	-4.025*** (0.461)	-3.905*** (0.628)	-5.917*** (1.309)
Implied η	1.386	1.808	5.025	4.905	6.917
Importer-species FE	Y	Y	Y	Y	Y
Exporter-importer FE	Y	Y	Y	Y	Y
Sample	Full	Full	Full	HHI High	HHI Low
N	8,760	6,223	6,223	2,988	3,153
First stage			-1.903*** (0.129)	-1.771*** (0.182)	-1.262*** (0.205)
CD-F			220.22	94.22	35.21

Notes: The Herfindahl–Hirschman Index is constructed to measure the concentration of exporters across species, such that $HHI_s = \sum_{i \in I} EXP_{i,s}^2$ where $EXP_{i,s}$ is the output share of country i from total output for species s . “HHI High” restricts the sample to the species of which the Herfindahl–Hirschman Index is above the median, while “HHI Low” restricts the sample to the species of which the Herfindahl–Hirschman Index is below the median.

The argument for the valid instrument is as follows. The relevance condition requires that the producer prices at the border for species s in exporter i are lower than that in the other countries if exporter i is located closer to the suitable habitat for species s than other countries are. The exclusion restriction holds as long as the distance to the suitable habitat is not correlated with the unobserved demand shifter.

Table 2 shows the estimates of η . Columns (1) and (2) show the estimation results from PPML and OLS, respectively. Column (3) shows the result from IV regression, where I obtain the elasticity of substitution across exporters $\eta = 5.025$. The first stage is negative, confirming the relevance condition that the proximity to suitable habit is associated with lower fishery prices. The large difference in the coefficient between OLS and IV indicates that the endogeneity concern from OLS could be severe.

A concern for exclusion restriction is that tastes toward the importer could reflect the proximity to suitable habitats. For example, consumers might have built up a particular taste for Norwegian Salmon since Norway is known to have a better habitat for salmon. If such concern is valid, the coefficient might have an upward bias since the preference shifter would be positively correlated with the proximity to suitable habitats. To investigate the validity of the concern, I estimate the parameter restricting the sample by whether the species are produced by multiple countries. In particular, I construct the relevant Herfindahl–Hirschman

Index to measure the concentration of exporters across species such that

$$HHI_s = \sum_{i \in I} EXP_{i,s}^2$$

where $EXP_{i,s}$ is the output share of country i in total output of species s .

Column (4) restricts the sample to the species of which the production is concentrated in particular countries such that the HHI is above the median across all species. While the point estimate of column (4) is larger than that of column (3), I do not find a statistical significance that the coefficients are different across samples. Column (5) restricts the sample to the species of which the production is dispersed across countries such that the HHI is below the median across all species. While the point estimate of column (5) is smaller than that of column (3), I again do not find a statistical significance that the coefficients are different across samples.

5.1.2 Elasticity of Substitution across Species (ν)

Next, I estimate the elasticity of substitution across species ν from the gravity equation, in the similar step from the previous subsection.

$$\log \left(\frac{X_{j,s}}{X_j} \right) = (1 - \nu) \log P_{j,s} + \phi_j + \varepsilon_{j,s}$$

where $X_{j,s}/X_j$ is the expenditure share for species s in all expenditure by importer j , $P_{j,s}$ is the consumer prices for fishery bundle of importer j for species s , ϕ_j is the importer fixed effect, and $\varepsilon_{j,s}$ is the residual term.

To construct the consumer prices for fishery bundle $P_{j,s}$, I use the observed producer prices at the border $p_{i,s}$ and the estimates of the preference shifter b_{ij} , the trade cost $d_{ij,s}$, and the elasticity of substitution across exporters η from the previous subsection such that

$$P_{j,s} = \left(\sum_i b_{ij} d_{ij,s}^{1-\eta} p_{i,s}^{1-\eta} \right)^{\frac{1}{1-\eta}}$$

Similarly, a concern about endogeneity arises because prices can be correlated with unobserved demand shifters. To alleviate this concern, I instrument the consumer prices $P_{j,s}$ with the proximity to suitable habitats $Z_{j,s}$ by importer j for species s . The relevance condition requires that the consumer prices in importer j for species s are lower than that for the other species if importer j is located closer to the suitable habitat of species s than to the other species. The exclusion restriction assumes that geography is uncorrelated with unobserved

Table 3: Estimation of ν

	(1)	(2)	(3)	(4)	(5)
	PPML	OLS	IV	IV	IV
$\log P_{js}$	-0.695*** (0.268)	-0.938*** (0.277)	-6.562*** (2.507)	-7.889*** (3.429)	-6.872*** (3.095)
Implied ν	1.695	1.938	7.562	8.889	7.872
Importer FE	Y	Y	Y	Y	Y
Sample	Full	Full	Full	HHI High	HHI Low
N	300	300	300	150	150
First stage			-1.155*** (0.394)	-1.289*** (0.543)	-0.798*** (0.215)
CD-F			11.14	4.49	11.30

Notes: The Herfindahl–Hirschman Index is constructed to measure the concentration of importers across species, such that $HHI_s = \sum_{j \in I} IMP_{j,s}^2$ where $IMP_{j,s}$ is the expenditure share of country j from total expenditure of species s . “HHI High” restricts the sample to the species of which the Herfindahl–Hirschman Index is above the median, while “HHI Low” restricts the sample to the species of which the Herfindahl–Hirschman Index is below the median.

demand shifters across species.

Table 3 shows the estimates for ν . Columns (1) and (2) show the estimation results from PPML and OLS. Column (3) shows the result from IV regression, where I obtain $\nu = 7.562$. The first stage is negative, as expected, confirming that the relevance condition holds. The large difference in the coefficient between OLS and IV justifies the need for the instrument.

Again, a concern for exclusion restriction is that tastes toward the species could reflect the proximity to suitable habitats. For instance, Norwegian consumers might have built up a particular taste for salmon if they were disproportionately exposed to salmon and lacked exposure to other species whose habitat is distant from Norway. The coefficient would be upward-biased if such concern is valid since the preference shifter is positively associated with the proximity to suitable habitats. To study the validity of the concern, I again estimate the parameter restricting the sample by whether the species are consumed by multiple countries. I construct the relevant Herfindahl–Hirschman Index to measure the concentration of importers across species such that:

$$HHI_s = \sum_{j \in I} IMP_{j,s}^2$$

where $IMP_{j,s}$ is the expenditure share of country j in total expenditure of species s .

Column (4) restricts the sample to the species of which the consumption is concentrated in particular countries such that the HHI is above the median across all species. Column (5) restricts the sample to the species of which the consumption is dispersed across countries such

that the HHI is below the median across all species. I do not find a statistical significance that the coefficients are different across samples.

5.1.3 Elasticity of Substitution across Locations (κ)

Lastly, I estimate the elasticity of substitution across locations κ from the indirect inference. The idea is that equation (2) about the relationship between the fuel subsidies and the location of fishing is informative about the elasticity of substitution across locations κ . The impact of fuel subsidies on the outputs at high sea relative to EEZ depends on how substitutable the varieties across regions are.

To estimate the elasticity of substitution across locations κ , I start by simulating the extended version of the model in Section 4.4 with the policy instruments. For a given κ , I shock the model with exogenous changes in the fuel subsidies over time that are calibrated to match the changes in fuel subsidies from actual data, where the fuel subsidies are modeled to target the high sea to generate a disproportionate effect on fishing location. Then, for each κ , I run equation (2) using the simulated data. Finally, I pick κ that generates the coefficient β_1 , which best matches the coefficient from Table 1 using the actual data. The detailed step is described in Appendix D.1.

Appendix Figure B.1 shows the relationship between the elasticity of substitution parameter across locations κ and the estimated coefficient β_1 , confirming that the effect of fuel subsidies on the outputs at high sea relative to EEZ is larger when the elasticity of substitution across locations is larger. From the indirect inference, I obtain the elasticity of substitution $\kappa = 8.08$.

5.2 Calibration of Remaining Parameters

This subsection calibrates the remaining parameters along with fundamentals. I set the stock elasticity of output $\xi = 0.79$, following Zhang (2011). The discount rate is set to $\beta = 0.97$, following Nordhaus (2007). As discussed in Section 4.3, the Pareto weights are calibrated to have equal weights across countries such that $\phi_i = 1$.

Taking the model to data in 2018, I use the standard model inversion methods to recover the fundamentals. For the demand shifter toward the fishery bundle b_i , I directly match the fishery expenditure for each country i by leveraging the property of quasi-linear preference. For the demand shifter toward the species b_s , I match the expenditure share of species s in total fishery expenditure. For the demand shifter toward the exporter b_{ij} , I match the expenditure share of exporter i in total expenditure by importer j . Lastly, for the demand

Table 4: Parameter calibration

Parameter	Value	Source
<i>a. Preferences</i>		
Demand shifter	$b_i, b_s, b_{ij}, b_{k,s}$	Model inversion
Elasticity of substitution	$\nu = 7.56, \eta = 5.03$	IV regression
	$\kappa = 8.08$	Subsidy regression
<i>b. Technology and geography</i>		
Productivity shifter	z_i^o, z_i	Model inversion
Stock elasticity	$\xi = 0.79$	Zhang (2011)
Commuting costs	$\tau_{ik,s}$	Model inversion
Trade costs	$d_{ij,s}$	Gravity residuals
<i>c. Planner-related parameter</i>		
Growth rate	$\gamma_{k,s}$	CMSY++ Package
Discount rate	$\beta = 0.97$	Nordhaus (2007)
Pareto weights	$\phi_i = 1$	

shifter toward the fishing location $b_{k,s}$, I match the expenditure share of fishing location k in all fishing locations for species s .

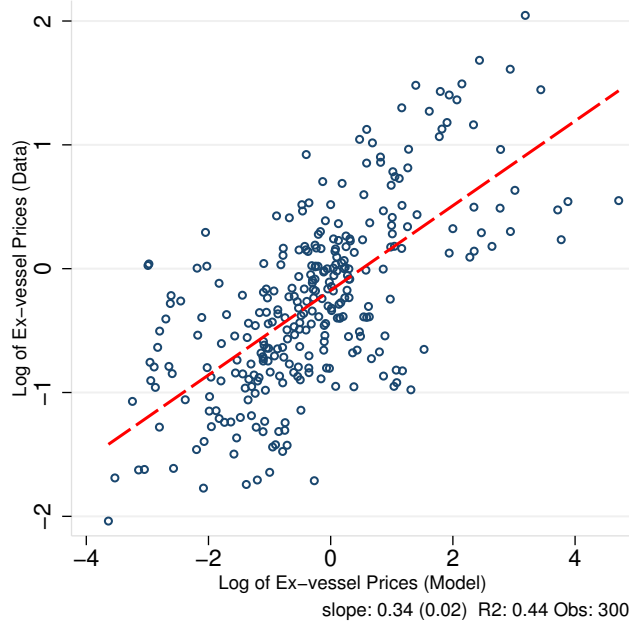
I then proceed to calibrate the productivity shifters. For the outside good productivity shifter z_i^o , I match the GDP per capita for each country i . For the fishery productivity shifter z_i , I match the levels of fishery output of country i . I recover the commuting costs $\tau_{ik,s}$ by matching the output share of fishing location k in all fishing locations by country i for species s . I use the gravity equation to recover the trade costs $d_{ij,s}$. Table 4 provides the summary of parameter calibration.

5.3 Model Fit

Before I present the quantitative results, I assess the model fit using non-targeted moments. While the calibration process targets the fishery outputs and expenditures, one of the non-targeted moments is the ex-vessel prices, which are the fishery prices when landed at the domestic port. As described in Subsection 5.1, these prices are directly observed in the data. In the model, I construct these prices by aggregating producer prices across fishing locations for each species and country following equation (26).

Figure 3 compares the ex-vessel prices implied by the model against the observed data. The observed data is normalized so that the mean of the observed ex-vessel prices equal to the mean of the ex-vessel prices implied by the model. The ex-vessel prices implied by the model fit well with the ex-vessel prices from the data, confirming the validity of the model.

Figure 4: Model Fit



Notes: The figure shows the relationship between the ex-vessel prices implied by the model (x-axis) and the observed data (y-axis), where the ex-vessel prices refer to the prices of fish when landed at the domestic port. The units of observation are at the country-species level. The model-implied ex-vessel prices are constructed by aggregating producer prices across fishing locations for each species and country, following equation (26). The ex-vessel prices from the data are normalized so that the average of ex-vessel prices from the data matches the average of ex-vessel prices from the model.

6 Quantitative Results

In this section, I examine the quantitative results from three scenarios: the decentralized equilibrium, the planner allocation, and a policy counterfactual. Using the model calibrated to data in 2018, I first project the economy forward under a Business As Usual (BAU) scenario to assess the trajectory of the decentralized equilibrium. I then contrast the BAU outcomes with those from the planner allocation to quantitatively illustrate the externalities from open access to global fishery. Finally, I simulate a policy counterfactual that permanently eliminates fuel subsidies across all countries. The difference in outcomes between the BAU and the counterfactual demonstrates the impact of fuel subsidies.

6.1 Decentralized Equilibrium (Business As Usual) and Planner

Table 5 summarizes the comparison of the outcomes between the BAU and the planner allocation. Under the BAU, the average stock to capacity ratio exhibits a substantial decline from 42.2% in the initial year to 28.6% at the steady state. When the planner internalizes the externalities of the fishery production, the average stock to capacity ratio increases to 53.8% at the steady state, representing an 88% increase relative to the BAU steady state.

Table 5: Quantitative Results: BAU and Planner

	BAU		Planner		% Difference	
	t=0	steady state	t=0	steady state	t=0	steady state
Average stock to capacity ratio	0.422	0.286	0.422	0.538	-	88.1%
Share of fishery labor					-35.1%	-39.0%
Share of outside good labor					0.28%	0.35%
Welfare from fishery					-7.8%	15.3%
Welfare from outside good					0.13%	0.12%
Welfare					-0.02%	0.39%
Net present value of welfare						0.11%

Notes: This table compares the outcomes from the BAU and the planner allocation for the initial period and the steady state. “% Difference” refers to the percentage difference between the BAU to the planner allocation for the corresponding period. Welfare from fishery refers to the welfare portion of fishery consumption in the quasi-linear preferences, while welfare from outside good refers to the welfare portion of outside good consumption in the quasi-linear preferences.

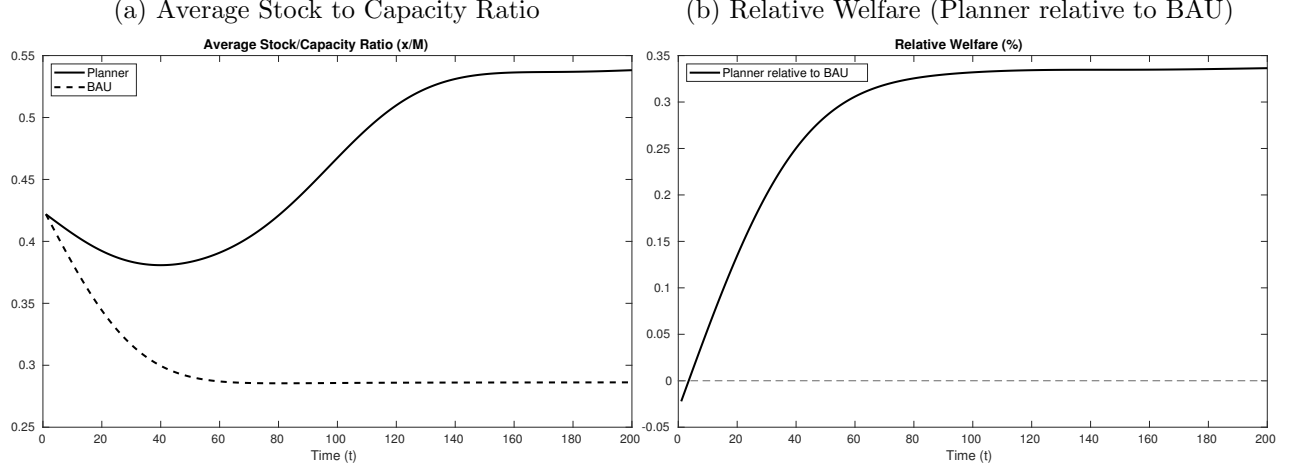
The socially optimum allocation by the planner involves initially relocating 35% of fishery labor to the outside good sector. This reallocation generates two immediate effects: a reduction in fishery output and an increase in outside good output. Given the quasi-linear utility preferences, I can decompose the welfare effects into fishery consumption and outside good consumption components. In the initial period, the welfare from fishery consumption decreases by 7.8%, while the welfare from outside good consumption increases by 0.13%.

The reallocation of fishery labor to the outside good sector facilitates stock accumulation over time. As stocks accumulate, the productivity of the fishery sector increases, allowing for larger outputs despite lower labor inputs. At the steady state, the welfare from fishery consumption increases by 15.3% in the planner allocation relative to the BAU. Additionally, the larger labor allocation to the outside good sector generates a 0.12% increase in the welfare from outside good consumption. These welfare gains from the two sectors result in a 0.39% increase in total global welfare at the steady state.

Figure 5 illustrates the dynamic transition paths under both scenarios. Panel (a) shows that even under the planner allocation, the average stock to capacity ratio initially decreases. This is because the planner, who discounts the future, prioritizes present consumption over future consumption. Panel (b) reveals the evolution of relative global welfare under the planner allocation compared to the BAU. While the trajectory shows a modest initial decline due to reduced fishery consumption, the increased fishery productivity from accumulated stocks generates a larger relative welfare at the steady state.

To evaluate the overall welfare implications, I construct the measure of the net present value of global welfare in the decentralized equilibrium following equations (20) and (21), which is comparable to the planner allocation. Accounting for the entire dynamic transition path, Table 5 shows that the net present value of global welfare under the planner allocation

Figure 5: Dynamic Path of Outcomes: BAU and Planner



Notes: Panel (a) shows the dynamic path of the average stock to capacity ratio for the BAU and the planner allocation. Panel (b) represents the welfare of the planner allocation relative to the BAU in percentage change.

increases by 0.11% relative to BAU.

We now turn to the spatial heterogeneity across fishing locations. Figure 6 shows the difference in the average stock to capacity ratio between the initial year and the steady states of the BAU and the planner allocation. Panel (a) illustrates the average stock to capacity ratio at the BAU steady state relative to the initial year. Compared to the the initial year, most fishing locations experience substantial declines in their average stock to capacity ratio at the BAU steady state, with the exception of a few fishing locations such as the EEZs of Chile and Bangladesh. The decline is more pronounced in the high seas compared to the EEZs, with the Central Pacific Ocean showing the most severe reduction in the average stock to capacity ratio.²⁴

Panel (b) depicts the average stock to capacity ratio at the steady state of planner allocation relative to the initial year. In contrast to the BAU scenario, as the planner internalizes the externalities of fishery production, many fishing locations show improved average stock to capacity ratios compared to their 2018 levels. The most evident contrast is that high seas show increases in their stock while most EEZs continue to experience declines.²⁵

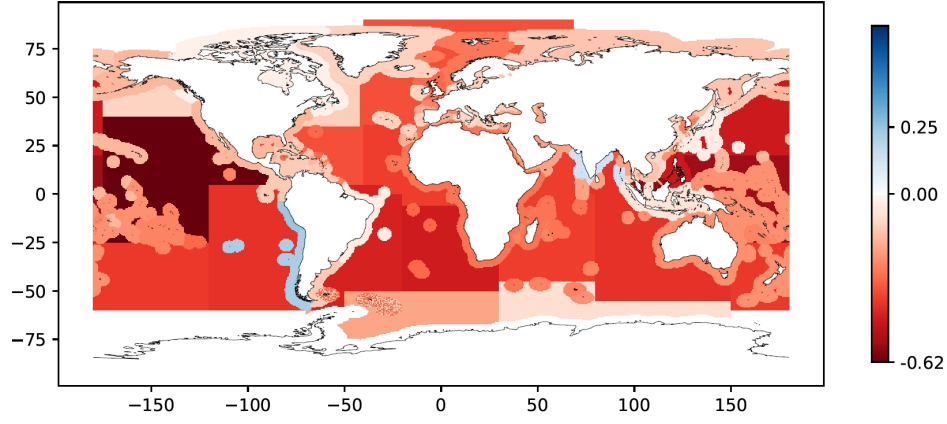
Panel (c) shows the average stock to capacity ratio at the steady state of planner allocation relative to the BAU steady state, providing the most direct comparison. Most fishing locations show a higher stock to capacity ratio except the EEZs in the African countries.

²⁴Appendix Figure B.3 panel (a) and (b) show the average stock to capacity ratio in the initial year and at the steady state under the BAU.

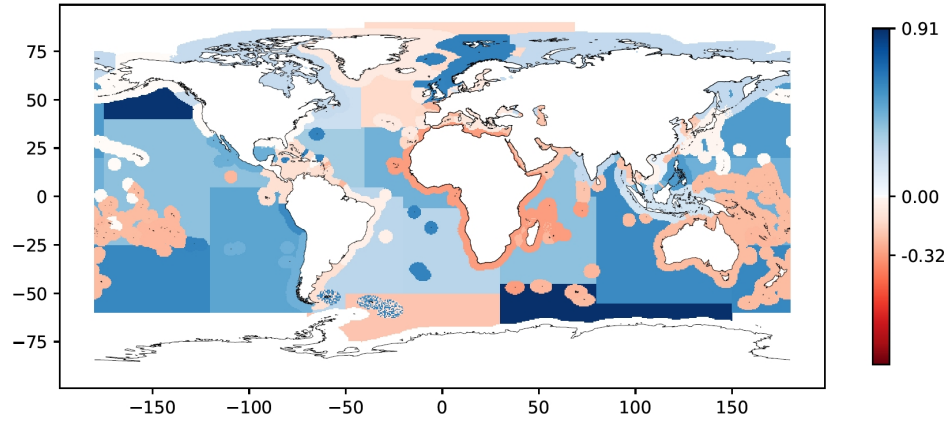
²⁵Appendix Figure B.3 panel (c) displays the average stock to capacity ratio at the steady state of the planner allocation.

Figure 6: Average Stock to Capacity Ratio (BAU and Planner)

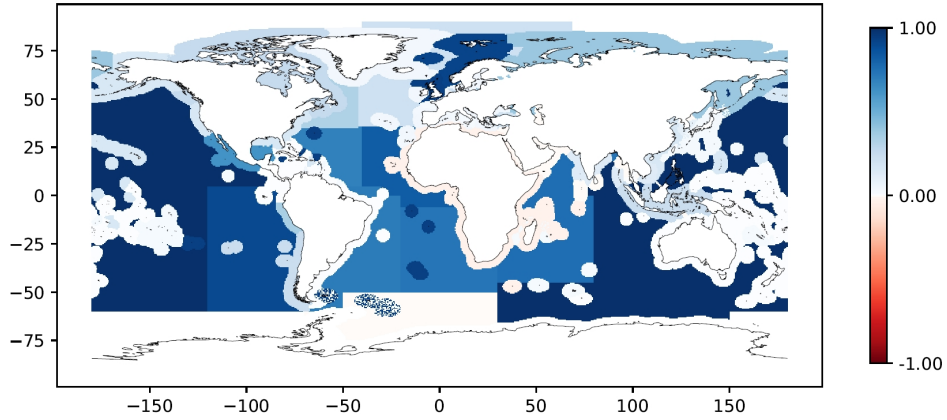
(a) BAU Steady State relative to Initial Year



(b) Planner Steady State relative to Initial Year



(c) Planner Steady State relative to BAU Steady State



Notes: The maps display the difference in average stock to capacity ratio across species in each fishing location from the quantitative exercises. Panel (a) shows the difference in average stock to capacity ratio between the initial year and the steady state under the BAU. Panel (b) depicts the difference in average stock to capacity ratio between the initial year and the steady state under the planner allocation. Panel (c) represents the difference in average stock to capacity ratio between the steady states under the BAU and the planner allocation.

Table 6: Quantitative Results: BAU and Counterfactual

	BAU		Counterfactual		% Difference	
	t=0	steady state	t=0	steady state	t=0	steady state
Average stock to capacity ratio	0.422	0.286	0.422	0.295	-	3.22%
<i>high sea</i>	0.379	0.090	0.379	0.099	-	10.3%
<i>EEZ</i>	0.431	0.326	0.431	0.335	-	2.76%
Welfare from fishery					-0.17%	1.42%
Welfare from outside good					0.001%	0.001%
Welfare					0.000%	0.01%
Net present value of welfare						0.004%

Notes: This table compares the outcomes from the BAU and the counterfactual for the initial period and the steady state. “% Difference” refers to the percentage difference between the BAU to the counterfactual for the corresponding period. Welfare from fishery refers to the welfare portion of fishery consumption in the quasi-linear preferences, while welfare from outside good refers to the welfare portion of outside good consumption in the quasi-linear preferences.

The most drastic increases occur in the high seas of the Pacific and the Indian Oceans, where the difference in average stock to capacity ratio is almost one, implying that fishery stocks recover from near depletion to approach the carrying capacity.

6.2 Policy Counterfactual: Eliminating Fuel Subsidies

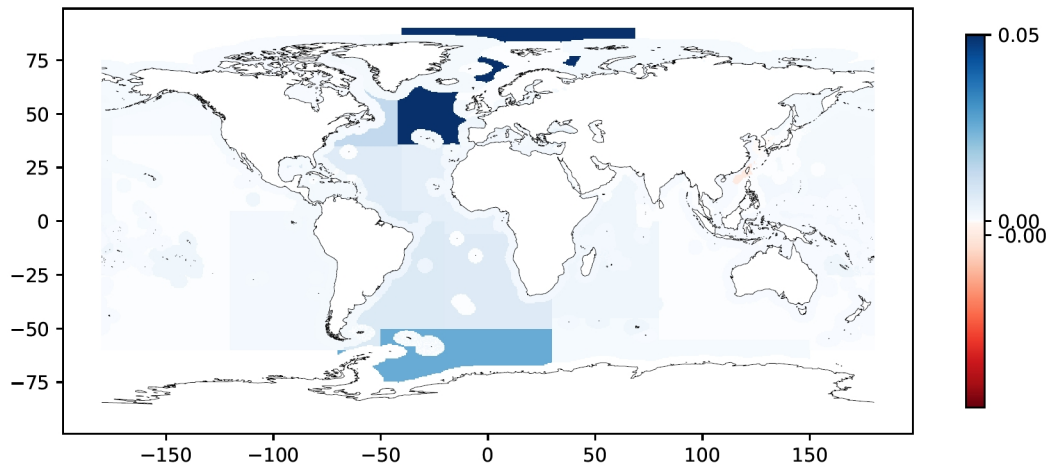
Next, I conduct a counterfactual analysis to examine the impact of fuel subsidies. Fuel subsidies have emerged as a major regulatory concern since they enable fishing fleets to exploit resources in remote high seas that would otherwise be economically inaccessible (Sumaila et al., 2019). Building on Pattern 3 in Section 3, which demonstrates the disproportionate effect of fuel subsidies on high seas output, I assume that fuel subsidies are exclusively targeted to high seas.

To counteract the fuel subsidies that are present in the BAU, I impose a production tax particularly on the high seas. From the extended model with policy instruments presented in Section 4.4, I calibrate country-specific tax to match the tax revenues in the counterfactual with the observed fuel subsidies in the data.

Table 6 compares the outcomes between the BAU and the counterfactual where the fuel subsidies are eliminated. While the average stock to capacity ratio falls to 28.6% at the steady state under the BAU, the ratio decreases only to 29.5% in the counterfactual. This indicates that the fuel subsidies reduce the average stock to capacity ratio by 3.22% at the steady state.

Given that fuel subsidies exclusively target fishing at the high seas, their impact exhibits significant heterogeneity between the high seas and the EEZs. In the BAU scenario, the high seas experience a dramatic decline in the average stock to capacity ratio from 37.9%

Figure 7: Average Stock to Capacity Ratio (Counterfactual SS relative to BAU SS)



Notes: The map shows the difference in the average stock to capacity ratio between BAU steady state and counterfactual steady state.

initially to 9.0% at steady state. In contrast, the EEZs show a more moderate decline from 43.1% to 32.6%, suggesting that resource depletion predominantly occurs in the high seas. Eliminating fuel subsidies increases the average stock to capacity ratio to 9.9% in high seas and 33.5% in EEZs, representing increases of 10.3% and 2.76%, respectively, confirming the disproportionate impact on high seas.

The elimination of fuel subsidies initially raises the price of the fisheries and decreases the demand for the fishery, reallocating labor to the outside good, which is a pattern consistent with the planner allocation. While this reallocation of labor reduces the welfare from fishery consumption initially, fishing productivity increases in the long run as fishery stock accumulates. At the steady state, despite lower labor inputs, welfare from fishery consumption exceeds the BAU level. Accounting for the entire dynamic path, the net present of welfare increases by 0.004% in the counterfactual relative to the BAU, indicating that fuel subsidies generate negative welfare impacts globally.²⁶

Figure 7 illustrates the spatial distribution of changes in the average stock to capacity ratio between the counterfactual and BAU steady state across fishing locations. The results reveal substantial spatial heterogeneity in the impact of eliminating fuel subsidies. Most EEZs demonstrate minimal increase in stock or even experience lower stock as in Taiwan's EEZ. In contrast, high seas exhibit substantial gains, particularly throughout the Atlantic Ocean, with the Northeast Atlantic experiencing the most drastic increase.

²⁶Appendix Figure B.2 presents the dynamic path of the outcomes for the counterfactual and the BAU.

7 Conclusions

This paper develops a dynamic spatial model of global fisheries to quantify the externalities arising from open access and evaluate the impact of fishery policies on global welfare. I calibrate the model and estimate key elasticities by leveraging novel geospatial data on fishery catch and stocks, providing a comprehensive framework for understanding global fishery.

The results reveal substantial inefficiencies under the Business As Usual scenario of the decentralized equilibrium. Under open access to the fishery, the atomistic firms fail to account for the externalities from fishery production, leading to the overexploitation of fishery stocks. In contrast, the social planner fully internalizes the externalities and reallocates the labor from the fishery sector to the outside good sector. While reducing the short-term fishery output, this reallocation leads to long-term gains through larger stock and increased productivity. The stark difference between the BAU and the planner allocation underscores the magnitude of externalities in global fisheries and the potential gains from policy interventions.

The counterfactual analysis of eliminating fuel subsidies yields two policy insights. First, these subsidies generate negative welfare effects at the global level, with their elimination leading to both higher stock levels and increased welfare. This highlights the importance of multilateral coordination in global fishery policy. Second, while eliminating fuel subsidies is helpful, the resulting stock levels and welfare outcomes remain substantially below the social optimum. This indicates that more comprehensive policy instruments may be necessary to fully internalize the externalities from open access to global fishery.

It is crucial to acknowledge the feasibility of implementing the socially optimal allocation. To implement the socially optimal allocation, the corresponding optimal policy should vary across exporters, importers, fishing locations, species, and time. In practice, the policy tools are more constrained. For instance, production policies often lack the dimension of importers, while trade policies may not account for the dimension of fishing locations.

The contingent trade policies described in [Harstad \(2024\)](#) could offer a feasible approach. Such policies could mandate the specification of fishery fishing locations in trade agreements, with tariff rates contingent on the gap between current and optimal stock. For example, tariffs could be imposed when stocks fall below optimal levels, while subsidies could be offered when stocks exceed optimal levels. This paper provides theoretical foundations and quantitative evidence to support the design and implementation of such policies.

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A Data Details

List of Country. Appendix Table A.1 provides the mapping of individual country ISO to aggregated 30 regions in the quantitative model.

Table A.1: List of Countries

Name	Region	Name	Region	Name	Region
Bangladesh	BGD	Algeria	XAF	Kuwait	XAS
Brazil	BRA	Liberia	XAF	Cambodia	XAS
Canada	CAN	Mali	XAF	Iraq	XAS
Chile	CHL	Mauritius	XAF	Kyrgyz Republic	XAS
China	CHN	Zimbabwe	XAF	United Arab Emirates	XAS
Germany	DEU	Senegal	XAF	Jordan	XAS
Spain	ESP	Mauritania	XAF	Saudi Arabia	XAS
France	FRA	Guinea	XAF	Singapore	XAS
United Kingdom	GBR	South Africa	XAF	Afghanistan	XAS
Indonesia	IDN	Burkina Faso	XAF	Sri Lanka	XAS
India	IND	Guyana	XAM	Armenia	XAS
Italy	ITA	Virgin Islands	XAM	Kazakhstan	XAS
Japan	JPN	Suriname	XAM	Timor-Leste	XAS
Korea, Rep.	KOR	Aruba	XAM	Bhutan	XAS
Mexico	MEX	Sint Maarten	XAM	Brunei Darussalam	XAS
Malaysia	MYS	St. Vincent and the Grenadines	XAM	Belarus	XEU
Norway	NOR	Barbados	XAM	Bosnia and Herzegovina	XEU
Peru	PER	Dominican Republic	XAM	Sweden	XEU
Philippines	PHL	Honduras	XAM	Estonia	XEU
Russian Federation	RUS	Bermuda	XAM	Slovenia	XEU
Thailand	THA	Trinidad and Tobago	XAM	Croatia	XEU
Turkey	TUR	Haiti	XAM	Faroe Islands	XEU
Taiwan	TWN	Bolivia	XAM	Greece	XEU
United States	USA	Paraguay	XAM	Iceland	XEU
Vietnam	VNM	Argentina	XAM	Montenegro	XEU
Mozambique	XAF	Antigua and Barbuda	XAM	San Marino	XEU
Tunisia	XAF	Cayman Islands	XAM	Romania	XEU
Cabo Verde	XAF	Panama	XAM	Netherlands	XEU
Chad	XAF	Dominica	XAM	Luxembourg	XEU
Burundi	XAF	Grenada	XAM	Slovak Republic	XEU
Congo, Dem. Rep.	XAF	Curacao	XAM	Andorra	XEU
Zambia	XAF	Colombia	XAM	Austria	XEU
Somalia	XAF	Belize	XAM	Switzerland	XEU
Sao Tome and Principe	XAF	St. Lucia	XAM	Ireland	XEU
Egypt, Arab Rep.	XAF	El Salvador	XAM	Isle of Man	XEU
Kenya	XAF	St. Kitts and Nevis	XAM	Ukraine	XEU
Rwanda	XAF	Puerto Rico	XAM	Monaco	XEU
Ghana	XAF	Jamaica	XAM	Malta	XEU
Nigeria	XAF	Nicaragua	XAM	Lithuania	XEU
Gambia	XAF	Greenland	XAM	Czech Republic	XEU
Morocco	XAF	Guatemala	XAM	Poland	XEU
Cameroon	XAF	Venezuela	XAM	Finland	XEU
Libya	XAF	Uruguay	XAM	Albania	XEU
Niger	XAF	Bahamas, The	XAM	North Macedonia	XEU
Guinea-Bissau	XAF	Ecuador	XAM	Serbia	XEU
Equatorial Guinea	XAF	Cuba	XAM	Bulgaria	XEU
Central African Republic	XAF	Costa Rica	XAM	Hungary	XEU
South Sudan	XAF	Turks and Caicos Islands	XAM	Belgium	XEU
Congo, Rep.	XAF	Yemen, Rep.	XAS	Liechtenstein	XEU
Cote d'Ivoire	XAF	West Bank and Gaza	XAS	Portugal	XEU
Uganda	XAF	Qatar	XAS	Denmark	XEU
Eswatini	XAF	Hong Kong	XAS	Latvia	XEU
Malawi	XAF	Iran, Islamic Rep.	XAS	Moldova	XEU
Comoros	XAF	Cyprus	XAS	Marshall Islands	XOC
Lesotho	XAF	Macao	XAS	American Samoa	XOC
Tanzania	XAF	Lebanon	XAS	Fiji	XOC
Sudan	XAF	Georgia	XAS	Vanuatu	XOC
Namibia	XAF	Mongolia	XAS	Nauru	XOC
Madagascar	XAF	Maldives	XAS	Northern Mariana Islands	XOC
Sudan	XAF	Turkmenistan	XAS	Micronesia, Fed. Sts.	XOC
Gabon	XAF	Nepal	XAS	Palau	XOC
Botswana	XAF	Pakistan	XAS	Kiribati	XOC
Benin	XAF	Oman	XAS	Papua New Guinea	XOC
Seychelles	XAF	Tajikistan	XAS	Tuvalu	XOC
Angola	XAF	Lao PDR	XAS	Samoa	XOC
Sierra Leone	XAF	Myanmar	XAS	New Zealand	XOC
Ethiopia	XAF	Bahrain	XAS	Guam	XOC
Togo	XAF	Israel	XAS	Solomon Islands	XOC
Djibouti	XAF	Uzbekistan	XAS	Australia	XOC
		Azerbaijan	XAS	Tonga	XOC

Notes: This table provides the mapping of individual country ISO (column ISO) to 30 regions (column region) in the quantitative model, which are 25 individual countries and 5 continental aggregates. The regions beginning with “X” refer to the continental aggregates. “XAF” stands for “Rest of Africa”, “XAM” stands for “Rest of America”, “XAS” stands for “Rest of Asia”, “XEU” stands for “Rest of Europe”, and “XOC” stands for “Rest of Oceania”.

List of Fishing Location. Appendix Table A.2 provides the list of 46 fishing locations in the quantitative model.

Table A.2: List of Fishing Location

EEZ				High Sea	
BGD	FRA	MEX	TWN	Atlantic, Northwest	Indian, Eastern
BRA	GBR	MYS	USA	Atlantic, Northeast	Indian, Antarctic
CAN	IDN	NOR	VNM	Atlantic, Western Central	Pacific, Northwest
CHL	IND	PER	XAF	Atlantic, Eastern Central	Pacific, Northeast
CHN	ITA	PHL	XAM	Atlantic, Southwest	Pacific, Western Central
DEU	JPN	RUS	XAS	Atlantic, Southeast	Pacific, Eastern Central
ESP	KOR	THA	XEU	Atlantic, Antarctic	Pacific, Southwest
		TUR	XOC	Indian, Western	Pacific, Southeast

Notes: This table provides the list of 46 fishing locations in the quantitative model, which consist of 30 EEZs and 16 high seas.

List of Species. Appendix Table A.3 provides the list of 10 species, based on ISSCAAP classification.

Table A.3: List of Fishery Species

Cods, hakes, haddocks	Shrimps, prawns	Other Demersal and Pelagic fish
Crustaceans and mollusks	Squids, cuttlefishes, octopuses	Other Diadromous fish
Herrings, sardines, anchovies	Tunas, bonitos, billfishes	Other fish not identified
Salmons, trouts, smelts		

Notes: This table provides the list of 10 fishery species in the quantitative model. While the original ISSCAAP classifies the fishery into 51 species, but I aggregate them into 10 larger categories to harmonize with the fishery trade dataset.

B Additional Tables and Figures

Table B.4: Summary Statistics of Fishery Stock

Species	Carrying Cap. (mil. tonnes)	Stock (mil. tonnes)		Stock/Cap. (%)		Stock (% chg.)	Stock/Cap. (p.p chg.)
		1980	2018	1980	2018		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Cods, hakes, haddocks	220.6	111.9	98.0	50.7	44.4	-6.3	-12.4
Crustaceans and mollusks	38.3	28.4	17.8	74.2	46.5	-27.6	-37.3
Herrings, sardines, anchovies	454.3	289.6	179.2	63.8	39.4	-24.3	-38.1
Salmons, trouts, smelts	80.7	59.9	40.1	74.3	49.7	-24.6	-33.1
Shrimps, prawns	18.7	14.3	9.5	76.1	50.5	-25.7	-33.7
Squids, cuttlefishes, octopuses	27.6	22.9	11.1	83.0	40.1	-43.0	-51.7
Tunas, bonitos, billfishes	112.3	83.1	57.7	74.0	51.3	-22.6	-30.6
Other Demersal and Pelagic fish	551.0	301.3	272.0	54.7	49.4	-5.3	-9.7
Other Diadromous fish	91.7	38.4	37.8	41.9	41.2	-0.7	-1.6
Other fish not identified	218.4	118.8	115.9	54.4	53.0	-1.3	-2.5

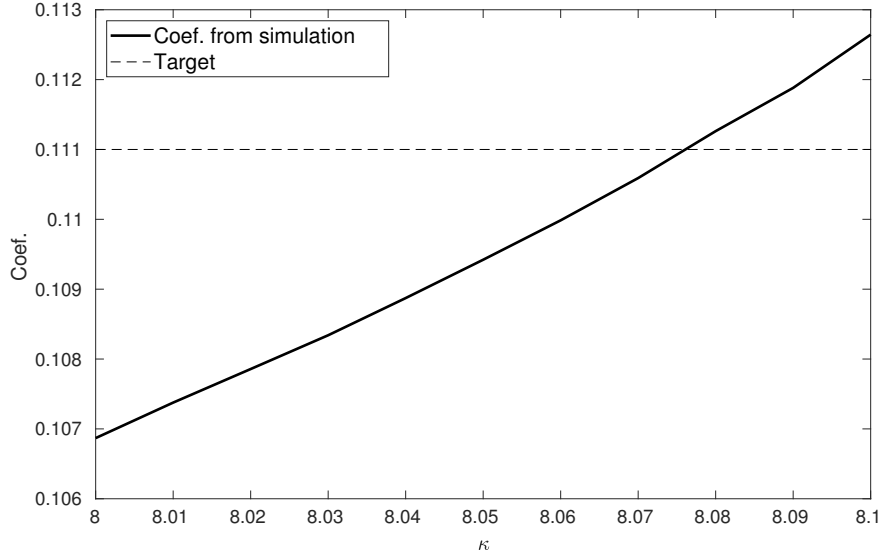
Notes: This table reports summary statistics of changes in global stock between 1980 and 2018 by fishery species used in quantitative model. Column 1 shows the global carrying capacity of species, where the global carrying capacity is calculated by summing the carrying capacity over all regions. Column 2 and 3 report the global stock of species in 1980 and 2018, where the global stock is calculated by summing the stock over all regions. Column 4 and 5 show the stock to capacity ratio in 1980 and 2018. Column 6 and 7 display the percentage change in stock and the percentage points change in stock to capacity ratio, respectively.

Table B.5: Summary Statistics of Fuel Subsidies

ISO	Fuel Subsidies in 2003 (MM)	Fuel Subsidies in 2018 (MM)	Country Share in 2003 (%)	Country Share in 2018 (%)
	(1)	(2)	(3)	(4)
CHN	1,814	3,433	28.85	44.44
JPN	1,115	632	17.73	8.18
KOR	331	604	5.26	7.82
TWN	120	593	1.91	7.67
XEU	147	479	2.34	6.21
THA	241	251	3.83	3.25
XAF	129	246	2.06	3.19
USA	243	201	3.86	2.60
IDN	171	141	2.72	1.83
XAS	116	121	1.84	1.56
XAM	129	113	2.05	1.47
ITA	0	104	0.00	1.35
IND	222	90	3.53	1.17
FRA	85	87	1.36	1.13
PHL	166	87	2.63	1.12
ESP	120	86	1.91	1.11
MYS	116	75	1.84	0.97
CAN	93	69	1.48	0.89
GBR	0	64	0.00	0.83
NOR	110	58	1.75	0.75
XOC	69	53	1.10	0.69
TUR	17	53	0.27	0.68
BGD	8	35	0.13	0.45
MEX	175	24	2.78	0.31
DEU	0	18	0.00	0.23
BRA	60	6	0.96	0.08
RUS	491	2	7.81	0.02
CHL	0	0	0.00	0.00
PER	0	0	0.00	0.00
VNM	0	0	0.00	0.00
World	6,288	7,727	100.00	100.00

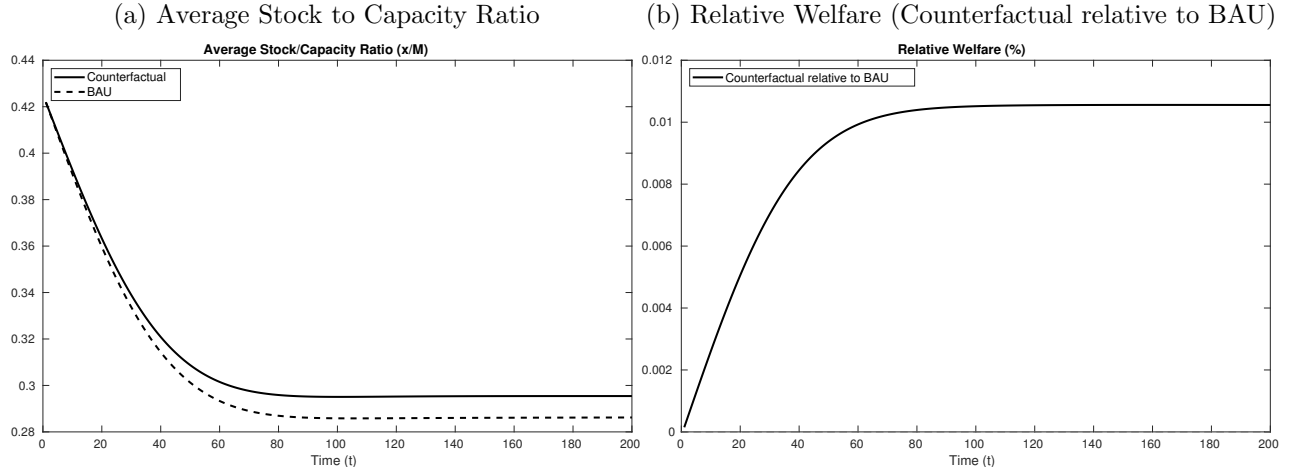
Notes: This table shows the summary statistics of fuel subsidies for our model regions. Column 1 and 2 report the fuel subsidies measured in million USD for 2003 and 2018, respectively. Column 3 and 4 report the country share of fuel subsidies among all regions in 2003 and 2018, respectively. The regions beginning with “X” refer to the continental aggregates. “XAF” stands for “Rest of Africa”, “XAM” stands for “Rest of America”, “XAS” stands for “Rest of Asia”, “XEU” stands for “Rest of Europe”, and “XOC” stands for “Rest of Oceania”.

Figure B.1: Relationship between κ and β_1



Notes: The figure shows the relationship between the parameter value of κ and the regression coefficient β_1 from equation (2).

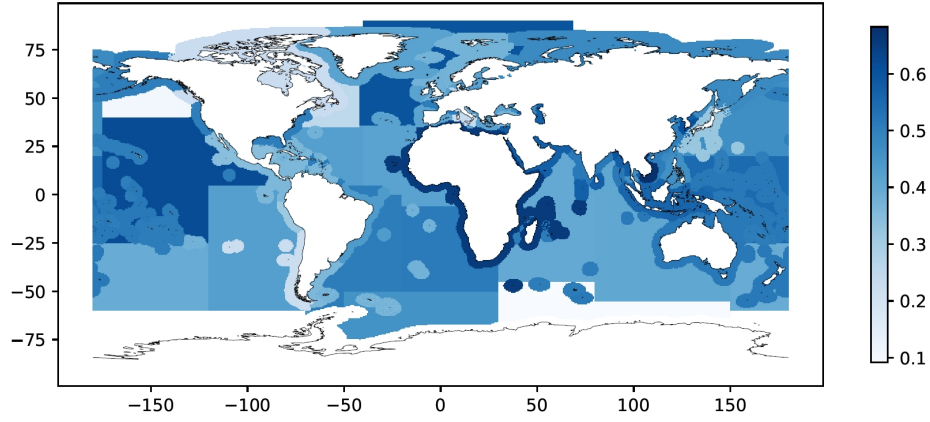
Figure B.2: Dynamic Path of Outcomes: BAU and Counterfactual



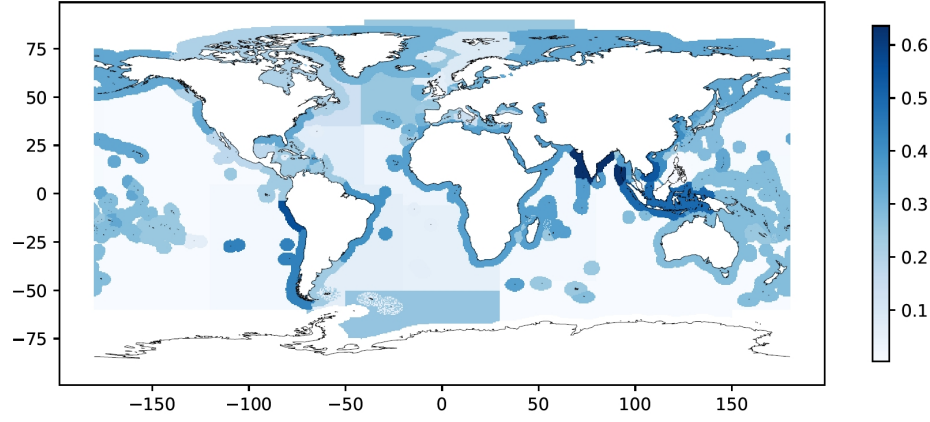
Notes: Panel (a) shows the dynamic path of the average stock to capacity ratio for both scenarios, the BAU and the counterfactual. Panel (b) represents the welfare from the counterfactual relative to the BAU, in percentage change.

Figure B.3: Average Stock to Capacity Ratio

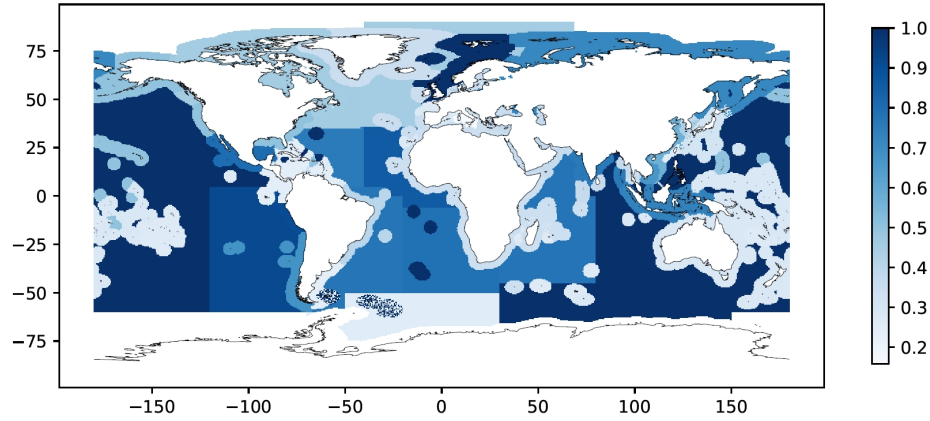
(a) Initial Year



(b) BAU Steady State



(c) Planner Steady State



Notes: The map shows the average stock to capacity ratio across species in each fishing location from the quantitative model. Panel (a) shows the average stock to capacity ratio in the initial year. Panel (b) and (c) represent the average stock to capacity ratio in the BAU steady state and the planner steady state, respectively.

C Derivations

C.1 Decentralized Equilibrium

- FOC of households

$$\max_{C_{ijk,st}, C_{j,t}^o} U_{j,t} \quad \text{s.t.} \quad \sum_{i \in I} \sum_{k \in K} \sum_{s \in S} p_{ijk,st} C_{ijk,st} + p_{j,t}^o C_{j,t}^o = E_{j,t} \quad \forall t$$

$$\begin{aligned} \mathcal{L} &= U_{j,t} + \lambda_{j,t} \left(E_{j,t} - p_{j,t}^o C_{j,t}^o - \sum p_{ijk,st} C_{ijk,st} \right) \\ [C_{j,t}^o] : \quad & \frac{\partial U_{j,t}}{\partial C_{j,t}^o} = \lambda_{j,t} p_{j,t}^o \\ [C_{ijk,st}] : \quad & \frac{\partial U_{j,t}}{\partial C_{ijk,st}} = \lambda_{j,t} p_{ijk,st} \end{aligned}$$

- Rearranging,

$$\frac{\partial U_{j,t}}{\partial C_{ijk,st}} = \frac{\partial U_{j,t}}{\partial C_{j,t}^o} \frac{p_{ijk,st}}{p_{j,t}^o} \quad (\text{C.1})$$

- FOC of atomistic fishery firms

$$\max_{L_{ijk,st}} \sum_t (p_{ijk,st} Q_{ijk,st} - w_{i,t} L_{ijk,st}) \quad \text{where} \quad Q_{ijk,st} = \frac{z_i (x_{k,st-1}/M_{k,s})^\xi}{d_{ij,s} \tau_{ik,s}} L_{ijk,st}$$

$$\begin{aligned} \mathcal{L} &= p_{ijk,st} Q_{ijk,st} - w_{i,t} L_{ijk,st} \\ [L_{ijk,st}] : \quad & p_{ijk,st} \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} = w_{i,t} \end{aligned} \quad (\text{C.2})$$

- Price

$$p_{ijk,st} = \frac{d_{ij,s} \tau_{ik,s}}{(x_{k,st-1}/M_{k,s})^\xi} \frac{w_{i,t}}{z_i} \quad (\text{C.3})$$

- FOC of outside good firms

$$\begin{aligned} \mathcal{L} &= p_{i,t}^o Q_{i,t}^o - w_{i,t}^o L_{i,t}^o \\ [L_{i,t}^o] : \quad & p_{i,t}^o \frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} = w_{i,t}^o \end{aligned} \quad (\text{C.4})$$

- Note that outside good is freely traded $p_{i,t}^o = p_{j,t}^o$ and labor is perfectly mobile $w_{i,t} = w_{i,t}^o$

- Putting (C.2) and (C.4) into (C.1), we get (19)

$$\frac{\partial U_{j,t}}{\partial C_{ijk,st}} = \frac{\partial U_{j,t}}{\partial C_{j,t}^o} \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} / \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right)$$

- From (C.2), we get (12)

$$p_{ijk,st} = \frac{d_{ij,s} \tau_{ik,s}}{z_i (x_{k,st-1} / M_{k,s})^\xi} w_{i,t}$$

C.2 Planner

$$\begin{aligned} \mathcal{L} = \sum_t \beta^t & \left[W_t + \sum_{k \in K} \sum_{s \in S} \lambda_{k,st} (x_{k,st-1} + G_{k,st} - H_{k,st} - x_{k,st}) + \sum_{i \in I} \mu_{i,t} \left(\bar{L}_i - \sum_{s \in S} \sum_{k \in K} \sum_{j \in I} L_{ijk,st} - L_{i,t}^o \right) \right. \\ & \left. + \sum_{i \in I} \sum_{s \in S} \sum_{k \in K} \sum_{j \in I} \theta_{ijk,st} (Q_{ijk,st} - C_{ijk,st}) + \alpha_t \left(\sum_{i \in I} Q_{i,t}^o - \sum_{j \in I} C_{j,t}^o \right) \right] \end{aligned}$$

- FOCs

$$[C_{ijk,st}] : \quad \theta_{ijk,st} = \frac{\partial W_t}{\partial C_{ijk,st}} \quad (C.5)$$

$$[C_{j,t}^o] : \quad \alpha_t = \frac{\partial W_t}{\partial C_{j,t}^o} \quad (C.6)$$

$$[L_{ijk,st}] : \quad \theta_{ijk,st} = \left(\mu_{i,t} + \lambda_{k,st} \frac{\partial H_{k,st}}{\partial L_{ijk,st}} \right) / \left(\frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right) \quad (C.7)$$

$$[L_{i,t}^o] : \quad \alpha_t = \mu_{i,t} / \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} \right) \quad (C.8)$$

$$[x_{k,st}] : \quad \lambda_{k,st} = \beta \left[\lambda_{k,st+1} \left(1 + \frac{\partial G_{k,st+1}}{\partial x_{k,st}} - \frac{\partial H_{k,st+1}}{\partial x_{k,st}} \right) + \sum_{i \in I} \sum_{j \in J} \theta_{ijk,st+1} \frac{\partial Q_{ijk,st+1}}{\partial x_{k,st}} \right] \quad (C.9)$$

$$[\mu_{i,t}] : \quad \bar{L}_i = \sum_{s \in S} \sum_{k \in K} \sum_{j \in I} L_{ijk,st} + L_{i,t}^o \quad (C.10)$$

$$[\theta_{i,st}] : \quad Q_{ijk,st} = C_{ijk,st} \quad (C.11)$$

$$[\alpha_t] : \quad \sum_{i \in I} Q_{i,t}^o = \sum_{j \in I} C_{j,t}^o \quad (C.12)$$

$$[\lambda_{k,st}] : \quad x_{k,st-1} + G_{k,st} = H_{k,st} + x_{k,st} \quad (C.13)$$

- From (C.6) and (C.8),

$$\mu_{i,t} = \frac{\partial W_t}{\partial C_{j,t}^o} \frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} \quad (\text{C.14})$$

- Putting (C.5) and (C.14) into (C.7),

$$\frac{\partial W_t}{\partial C_{ijk,st}} \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} = \frac{\partial W_t}{\partial C_{j,t}^o} \frac{\partial Q_{ij,t}^o}{\partial L_{i,t}^o} + \lambda_{k,st} \frac{\partial H_{k,st}}{\partial L_{ijk,st}} \quad (\text{C.15})$$

- From (C.9) and (C.13),

$$\lambda_{k,st} = \beta \left[\lambda_{k,st+1} \left(1 + \frac{\partial G_{k,st+1}}{\partial x_{k,st}} - \frac{\partial H_{k,st+1}}{\partial x_{k,st}} \right) + \frac{\partial W_{t+1}}{\partial C_{ijk,st+1}} \frac{\partial Q_{ijk,st+1}}{\partial x_{k,st}} \right] \quad (\text{C.16})$$

- Let $\Omega_{k,st} = \partial W_{t+1} / \partial x_{k,st}$ and $\Theta_{k,st} = \partial x_{k,st+1} / \partial x_{k,st}$

$$\lambda_{k,st} = \beta [\lambda_{k,st+1} \Theta_{k,st} + \Omega_{k,st}]$$

- At any point of time t ,

$$\frac{\partial W_t}{\partial C_{ijk,st}} = \frac{\partial W_t}{\partial C_{j,t}^o} \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} / \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right) + \lambda_{k,st} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}} \quad (\text{C.17})$$

- At the steady state, $\lambda_{k,st} = \lambda_{k,st+1} = \lambda_{k,st+2} \dots$

$$\lambda_{k,st} = \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \quad (\text{C.18})$$

- Putting (C.18) into (C.15), and rearranging, we get (22)

$$\frac{\partial W_t}{\partial C_{ijk,st}} = \frac{\partial W_t}{\partial C_{j,t}^o} \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} / \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right) + \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}} \quad (\text{C.19})$$

where t is at the steady state

C.3 Dual Approach

- Now the government imposes per unit tax $t_{ijk,st}$ where $\tilde{p}_{ijk,st} = p_{ijk,st} + t_{ijk,st}$

- FOC of households

$$\begin{aligned}\mathcal{L} &= U_{j,t} + \lambda_{j,t} \left(E_{j,t} - p_{j,t}^o C_{j,t}^o - \sum \tilde{p}_{ijk,st} C_{ijk,st} \right) \\ [C_{j,t}^o] : \quad & \frac{\partial U_{j,t}}{\partial C_{j,t}^o} = \lambda_{j,t} p_{j,t}^o \\ [C_{ijk,st}] : \quad & \frac{\partial U_{j,t}}{\partial C_{ijk,st}} = \lambda_{j,t} \tilde{p}_{ijk,st}\end{aligned}$$

- Rearranging,

$$\frac{\partial U_{j,t}}{\partial C_{ijk,st}} = \frac{\partial U_{j,t}}{\partial C_{j,t}^o} \frac{\tilde{p}_{ijk,st}}{p_{j,t}^o} \quad (\text{C.20})$$

- FOC of fishery firms

$$\begin{aligned}\mathcal{L} &= (\tilde{p}_{ijk,st} - t_{ijk,st}) Q_{ijk,st} - w_{i,t} L_{ijk,st} \\ [L_{ijk,st}] : \quad & (\tilde{p}_{ijk,st} - t_{ijk,st}) \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} = w_{i,t} \\ \tilde{p}_{ijk,st} &= w_{i,t} \left(\frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right)^{-1} + t_{ijk,st} \\ &= p_{ijk,st} + t_{ijk,st}\end{aligned} \quad (\text{C.21})$$

- FOC of outside good firms

$$\begin{aligned}\mathcal{L} &= p_{i,t}^o Q_{i,t}^o - w_{i,t}^o L_{i,t}^o \\ [L_{i,t}^o] : \quad & p_{i,t}^o \frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} = w_{i,t}^o\end{aligned} \quad (\text{C.22})$$

- Balance of Budget

$$w_{i,t}^o L_{i,t}^o + w_{i,t} L_{i,t} + t_{ijk,st} Q_{ijk,st} = X_{i,t}^o + X_{i,t}$$

- Claim: the following optimal policy $t_{ijk,st}^*$ realizes the optimal allocation from equation (C.19)

$$t_{ijk,st}^* = \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}}$$

- With $\phi_j = 1$, note that

$$\frac{\partial W_t}{\partial C_{ijk,st}} = \frac{\partial U_{j,t}}{\partial C_{ijk,st}}, \quad \frac{\partial W_t}{\partial C_{j,t}^o} = \frac{\partial U_{j,t}}{\partial C_{j,t}^o}$$

- Then, (C.19) becomes

$$\frac{\partial U_{j,t}}{\partial C_{ijk,st}} = \frac{\partial U_{j,t}}{\partial C_{j,t}^o} \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} / \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right) + \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}}$$

- Plugging in (C.20), (C.21), and (C.22) into (C.19)

$$\frac{\partial U_{j,t}}{\partial C_{j,t}^o} \frac{\tilde{p}_{ijk,st}}{p_{j,t}^o} = \frac{\partial U_{j,t}}{\partial C_{j,t}^o} \frac{(\tilde{p}_{ijk,st} - t_{ijk,st}^*)}{w_{i,t}} + \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}}$$

- With the quasi-linear preferences, free labor mobility, and choice of numeraire where $\partial U_{j,t} / \partial C_{j,t}^o = 1$, $w_{i,t} = z_i^o$, and $p_{j,t}^o = 1$, the optimal policy $t_{ijk,st}^*$ is indeed

$$\begin{aligned} \tilde{p}_{ijk,st} &= (\tilde{p}_{ijk,st} - t_{ijk,st}^*) + \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}} \\ \Leftrightarrow t_{ijk,st}^* &= \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}} \end{aligned}$$

C.4 Proof for Lemma

Want to show: the sufficient conditions i) $\xi \geq 1$ or ii) $0 < \xi < 1$ and $\gamma_{k,s} < \frac{1}{1-\xi} \left(\frac{1}{\beta} - 1 \right)$ guarantee $\theta_{k,st} < 1/\beta$ for all ranges of $\frac{x_{k,st}}{M_{k,s}} \in [0, 1]$ at the steady state.

Taking partial derivative of equation (9) with respect to $x_{k,st}$,

$$\theta_{k,st} = \frac{\partial x_{k,st+1}}{\partial x_{k,st}} = \frac{\partial G_{k,st+1}}{\partial x_{k,st}} - \frac{\partial H_{k,st+1}}{\partial x_{k,st}} = (1 + \gamma_{k,s}) - 2\gamma_{k,s} \frac{x_{k,st}}{M_{k,s}} - \frac{\xi}{x_{k,s}} \sum_i \sum_j d_{ij,s} \tau_{ik,s} Q_{ijk,st} \quad (\text{C.23})$$

From equation (18), at the steady state,

$$\gamma_{k,s} x_{k,st} \left(1 - \frac{x_{k,st}}{M_{k,s}} \right) = \sum_{i \in I} \sum_{j \in I} d_{ij,s} \tau_{ik,s} Q_{ijk,st} \quad (\text{C.24})$$

Combining (C.23) and (C.24), at the steady state,

$$\begin{aligned} \theta_{k,st} &= (1 + \gamma_{k,s}) - 2\gamma_{k,s} \frac{x_{k,st}}{M_{k,s}} - \xi \gamma_{k,s} \left(1 - \frac{x_{k,st}}{M_{k,s}} \right) \\ &= 1 + \gamma_{k,s} \left(1 - 2 \frac{x_{k,st}}{M_{k,s}} - \xi \left(1 - \frac{x_{k,st}}{M_{k,s}} \right) \right) \end{aligned}$$

- 1) $\xi \geq 1$

For all ranges of $\frac{x_{k,st}}{M_{k,s}} \in [0, 1]$, I have

$$1 - 2\frac{x_{k,st}}{M_{k,s}} - \xi(1 - \frac{x_{k,st}}{M_{k,s}}) < 1$$

Thus,

$$\Theta_{k,st} < 1 < 1/\beta$$

$$2) \ 0 < \xi < 1 \text{ and } \gamma_{k,s} < \frac{1}{1-\xi} \left(\frac{1}{\beta} - 1 \right)$$

$$\begin{aligned} \gamma_{k,s} &< \frac{1}{1-\xi} \left(\frac{1}{\beta} - 1 \right) \\ \Leftrightarrow \left(1 - \frac{1}{2-\xi} \right) - \frac{1}{(2-\xi)\gamma_{k,s}} \left(\frac{1}{\beta} - 1 \right) &< 0 \end{aligned} \tag{C.25}$$

At the steady state,

$$\begin{aligned} \Theta_{k,st} = 1 + \gamma_{k,s} \left(1 - 2\frac{x_{k,st}}{M_{k,s}} - \xi(1 - \frac{x_{k,st}}{M_{k,s}}) \right) &< 1/\beta \\ \Leftrightarrow \frac{x_{k,st}}{M_{k,s}} > \left(1 - \frac{1}{2-\xi} \right) - \frac{1}{(2-\xi)\gamma_{k,s}} \left(\frac{1}{\beta} - 1 \right) \end{aligned} \tag{C.26}$$

Combining (C.25) and (C.26),

$$\Theta_{k,st} < 1/\beta \quad \text{for all ranges of } \frac{x_{k,st}}{M_{k,s}} \in [0,1]$$

D Calibration Details

D.1 Steps for Calibration of κ

1. Under the environment described in 4.4, the government imposes the policy instruments $t_{ijk,st} = s_{i,t}$
2. For each κ , calibrate subsidy rate $s_{i,16}$ to match the change in amount of fuel subsidy assuming $s_{i,0} = 0$. That is, $\Delta_{0 \rightarrow 16} S_{i,t}^{model} = \Delta_{2002 \rightarrow 2018} S_{i,t}^{Data}$
3. Run regression using simulated data and obtain κ that matches β_1 from equation (2)

E Numerical Algorithm

E.1 Business As Usual

E.1.1 Static Equilibrium

For every period, given the geography, fundamentals, and endowment including fishery stock $x_{k,st}$, solve for the allocation of labor $L_{ijk,st}$

1. Initial guess for the allocation of labor $L_{ijk,st}$
2. Compute quantity and output
3. Compute prices and wages and expenditure
4. Compute $L_{ijk,st}^{new} = \frac{Y_{ijk,st}}{w_{i,t}}$
5. Update $L_{ijk,st} = \alpha L_{ijk,st}^{new} + (1 - \alpha) L_{ijk,st}$ and go back to Step 1, where $0 < \alpha < 1$ is the dampening parameter
6. Iterate until convergence

E.1.2 Steady State Equilibrium

1. Initial guess for steady state stock $x_{k,st}$
2. Given the guess of stock $x_{k,st}$, solve for static equilibrium as described in [E.1.1](#)
3. Compute the growth $G(x_{k,st})$ and harvest $H(x_{k,st})$ as
$$G(x_{k,st}) = \gamma_{k,s} x_{k,st} \left(1 - \frac{x_{k,st}}{M_{k,s}}\right) \text{ and } H(x_{k,st}) = \sum_{i \in I} \sum_{j \in I} d_{ij,s} \tau_{ik,s} Q_{ijk,st}$$
4. Update $x_{k,st} = x_{k,st} \times \left(\frac{G(x_{k,st})}{H(x_{k,st})}\right)^\alpha$ and go back to Step 1, where α is the dampening parameter
5. Iterate until convergence

E.1.3 Dynamic Equilibrium

1. Given the stock endowment at the initial period, $x_{k,s0}$, solve for static equilibrium as described in [E.1.1](#)
2. Obtain $x_{k,s1}$ using the law of motion for fish

$$x_{k,st} - x_{k,st-1} = \gamma_{k,s} x_{k,st-1} \left(1 - \frac{x_{k,st-1}}{M_{k,s}}\right) - \sum_{i \in I} \sum_{j \in I} d_{ij,s} \tau_{ik,s} Q_{ijk,st}$$

3. Solve for static equilibrium as described in [E.1.1](#)
4. Solve for $t = 2, \dots, T$

E.2 Planner's Problem

E.2.1 Steady State Equilibrium

1. Initial guess for steady state stock $x_{k,st}$
2. Given $x_{k,st}$, solve for $L_{ijk,st}$ that satisfies

$$\frac{\partial W_t}{\partial C_{ijk,st}} = \frac{\partial W_t}{\partial C_{i,t}^o} \left(\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} / \frac{\partial Q_{ijk,st}}{\partial L_{ijk,st}} \right) + \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}} \frac{\partial H_{k,st}}{\partial Q_{ijk,st}}$$

where $\Omega_{k,st} = \frac{\partial W_t}{\partial x_{k,st}}$ and $\Theta_{k,st} = \frac{\partial x_{k,st+1}}{\partial x_{k,st}}$

3. Compute the growth $G(x_{k,st})$ and harvest $H(x_{k,st})$
 where $G(x_{k,st}) = \gamma_{k,s} x_{k,st} (1 - \frac{x_{k,st}}{M_{k,s}})$ and $H(x_{k,st}) = \sum_{i \in I} \sum_{j \in I} d_{ij,s} \tau_{ik,s} Q_{ijk,st}$
4. Update $x_{k,st} = x_{k,st} \times \left(\frac{G(x_{k,st})}{H(x_{k,st})} \right)^\alpha$ and go back to step 1, where α is the dampening parameter
5. Iterate until convergence
6. Obtain the steady state $x_{k,st}, L_{ijk,st}, \lambda_{k,st}, \Omega_{k,st}, \Theta_{k,st}$ where $\lambda_{k,st} = \frac{\beta \Omega_{k,st}}{1 - \beta \Theta_{k,st}}$

E.2.2 Dynamic Equilibrium

Given the initial stock $x_{k,s0}$ and steady state $x_{k,st}, L_{ijk,st}, \lambda_{k,st}, \Omega_{k,st}, \Theta_{k,st}$ from [E.2.1](#), I assume that the last period converges to steady state such that $x_{k,sT} = x_{k,st}, L_{ijk,sT} = L_{ijk,st}$, and $\lambda_{k,sT} = \lambda_{k,st}$.

1. Initial guess for $\{x_{k,st}\}_{t=1}^{T-1}$ by interpolating $x_{k,st} = \frac{(T-t)x_{k,s0} + (t)x_{k,sT}}{T}$
2. Given $\{x_{k,st}\}_{t=0}^T$, roll backward from $t = T$
 - (a) solve backward for $\lambda_{k,st-1}$ using

$$\lambda_{k,st-1} = \beta [\lambda_{k,st} \Theta_{k,st} + \Omega_{k,st}]$$

(b) solve for $L_{ijk,t-1}$, $\Theta_{k,st-1}$, $\Omega_{k,st-1}$ using

$$\frac{\partial W_{t-1}}{\partial C_{ijk,st-1}} = \frac{\partial W_{t-1}}{\partial C_{i,t-1}^o} \left(\frac{\partial Q_{i,t-1}^o}{\partial L_{i,t-1}^o} / \frac{\partial Q_{ijk,st-1}}{\partial L_{ijk,st-1}} \right) + \lambda_{k,st-1} \frac{\partial H_{k,st-1}}{\partial Q_{ijk,st-1}}$$

(c) solve for $t = T - 2, T - 3, \dots, 1$ and obtain $\{L_{ijk,st}, \lambda_{k,st}, \Omega_{k,st}, \Theta_{k,st}\}_{t=1}^T$

3. Given $x_{k,s0}$ and $\{L_{ijk,st}, \lambda_{k,st}, \Omega_{k,st}, \Theta_{k,st}\}_{t=1}^T$, roll forward from $t = 1$

(a) solve for $x_{k,st}^{new}$ using the law of motion for fish

$$x_{k,st}^{new} - x_{k,st-1} = \gamma_{k,s} x_{k,st-1} \left(1 - \frac{x_{k,st-1}}{M_{k,s}} \right) - \sum_{i \in I} \sum_{j \in I} d_{ij,s} \tau_{ik,s} Q_{ijk,st}$$

4. Update $x_{k,st} = \alpha x_{k,st}^{new} + (1 - \alpha) x_{k,st}$ and go back to Step 1, where $0 < \alpha < 1$ is the dampening parameter

5. Iterate until convergence

E.3 Partial Derivatives

$$\frac{\partial W_t}{\partial C_{ijk,st}} = b_j b_s b_{ij} b_{k,s} C_{j,t}^{-1+\frac{1}{\nu}} C_{j,st}^{\frac{1}{\eta}-\frac{1}{\nu}} C_{ij,st}^{\frac{1}{\kappa}-\frac{1}{\eta}} C_{ijk,st}^{-\frac{1}{\kappa}}$$

$$\frac{\partial W_t}{\partial C_{j,t}^o} = 1$$

$$\frac{\partial Q_{ijk,kt}}{\partial L_{ijk,st}} = \frac{z_i}{d_{ij,s} \tau_{ik,s}} \left(\frac{x_{k,st-1}}{M_{k,s}} \right)^\xi$$

$$\frac{\partial Q_{i,t}^o}{\partial L_{i,t}^o} = z_i^o$$

$$\frac{\partial Q_{ijk,st}}{\partial x_{k,st-1}} = \frac{\xi}{x_{k,st-1}} z_i \left(\frac{x_{k,st-1}}{M_{k,s}} \right)^\xi L_{ijk,st}$$

$$\frac{\partial G_{k,st}}{\partial x_{k,st-1}} = (1 + \gamma_{k,s}) - 2\gamma_{k,s} \frac{x_{k,st-1}}{M_{k,s}}$$

$$\frac{\partial H_{k,st}}{\partial x_{k,st-1}} = \sum_i \sum_j d_{ij,s} \tau_{ik,s} \frac{\partial Q_{ijk,st}}{\partial x_{k,st-1}} = \sum_i \sum_j \frac{\xi}{x_{k,st-1}} z_i \left(\frac{x_{k,st-1}}{M_{k,s}} \right)^\xi L_{ijk,st} = \frac{\xi}{x_{k,st-1}} \sum_i \sum_j Q_{ijk,st}$$

$$\frac{\partial H_{k,st}}{\partial L_{ijk,st}} = d_{ij,s} \tau_{ik,s} \frac{\partial Q_{ijk,kt}}{\partial L_{ijk,st}}$$